

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 9

LECTURE 9

Introduction to Data-Driven Modeling

- Introduction
- System Identification Procedure
- Experiment Design & Data Examination
- Choice of Input Signal and Data Collection
- Input-Output Data Examination
- Linear Regression Modeling
 - Least-Squares Estimation

What We Already Know?

• Time Response of First-Order Systems

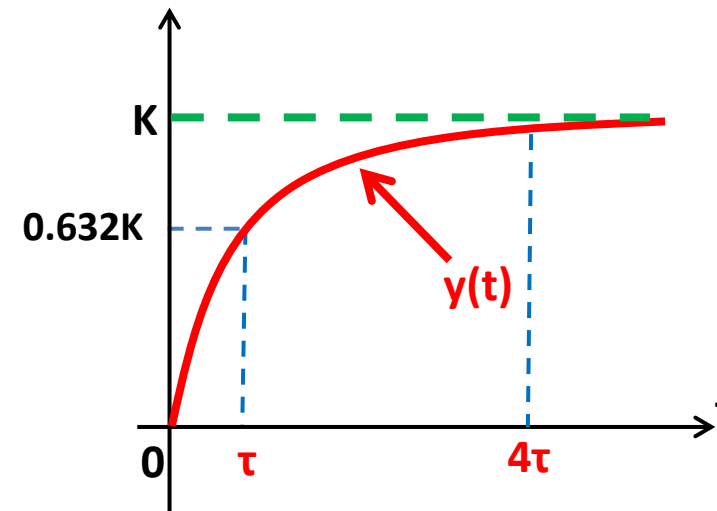
$$G(s) = \frac{K}{\tau s + 1}$$

K → Steady-state Gain

τ → Time Constant

Pole → $s = -\frac{1}{\tau}$

4τ → Settling-time



$$y(t) = K(1 - e^{-t/\tau}), \quad t \geq 0$$

• Time Response of Second-Order Systems

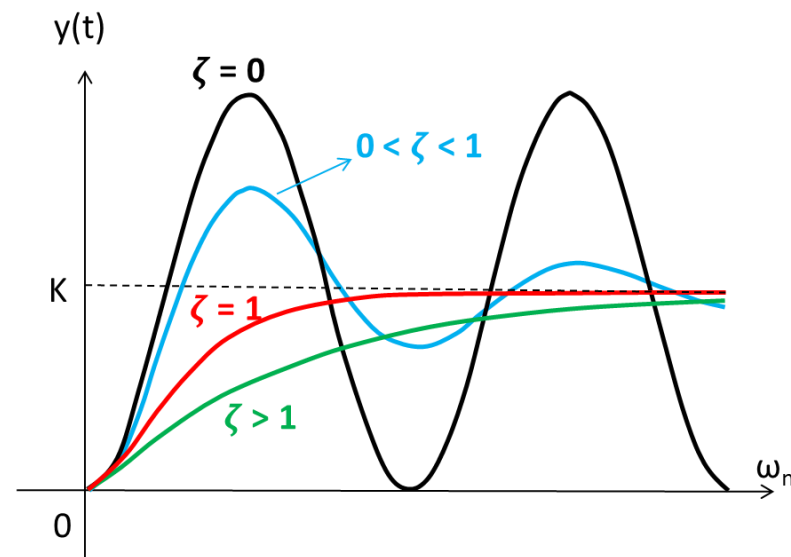
$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

K → Steady-state Gain

ζ → Damping ratio

ω_n → Undamped Natural Frequency

Poles → $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$



- **Stable Systems $\zeta > 0$**
 - Over-damped Systems $\zeta > 1$
 - Critically-damped Systems $\zeta = 1$
 - Under-damped systems $0 < \zeta < 1$
- **Marginally Stable Systems $\zeta = 0$**
 - Undamped Systems
- **Unstable Systems $\zeta < 0$**
 - Negatively-damped systems

What We Already Know?

- Time Response Specifications of Second-Order Systems

Rise time (t_r):

$$t_r \cong \frac{0.8 + 2.5\zeta}{\omega_n}$$

Peak time (t_p):

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

Maximum overshoot (M_p):

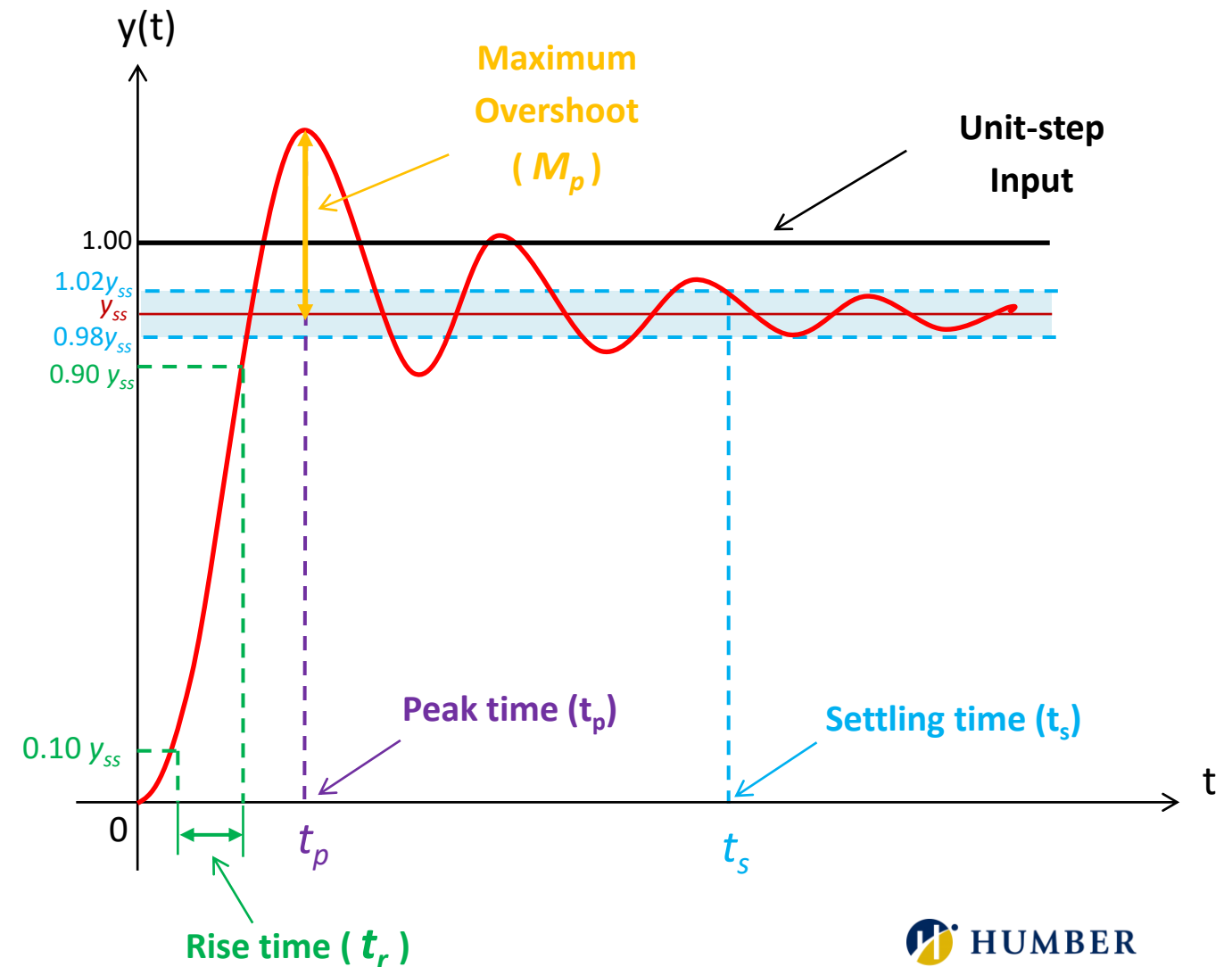
$$O.S. = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$\%O.S. = \frac{M_p}{y_{ss}} = \frac{y(t_p) - y_{ss}}{y_{ss}} \times 100\%$$

Settling time (t_s):

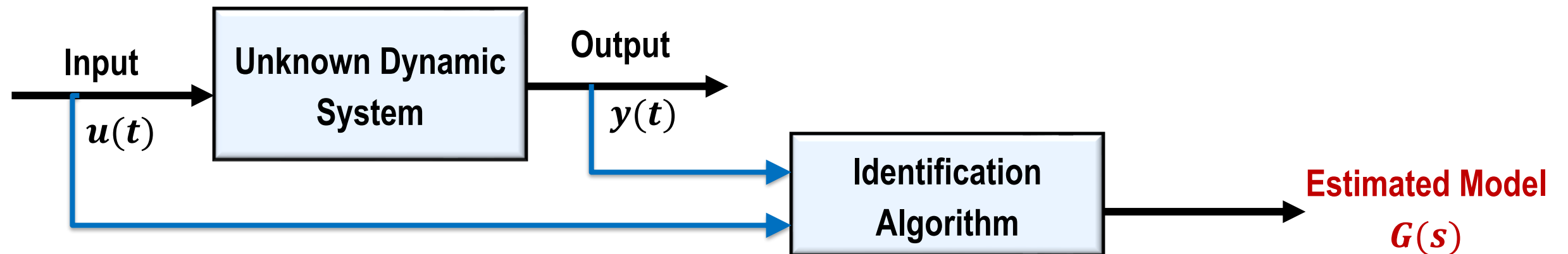
$$2\% \text{ criteria} \rightarrow t_s \approx \frac{4}{\zeta\omega_n}, \quad 0 < \zeta < 0.9$$

$$G(s) = \frac{K \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



Introduction to System Identification

- **System identification** refers to obtaining the **transfer function** $G(s)$ of a system by only considering its **output response** to a given particular **input signal**.
- This is useful when we have not much information about a system at hand.
- For example, it is a **black box** or **too complicated** to be modeled, and we need to find out its transfer function for simulation or control system design purposes.
- Although we assume that no information is provided by the system, some hypothesis must be considered, like for instance the **order** of the system.



- In this course we will focus on system identification and transfer function modeling via
 - **Transient response analysis** of **first-order** and **second-order** systems. (**Traditional graphical techniques**)
 - **Modern system identification methods** for **black-box** system identification. (**Statistical methods**)

System Modeling via Step Response

Example 1

Figure shows the response of a mass-spring-damper system to a step input of magnitude $6 \times 10^3 \text{ N}$.

The equation of motion is:

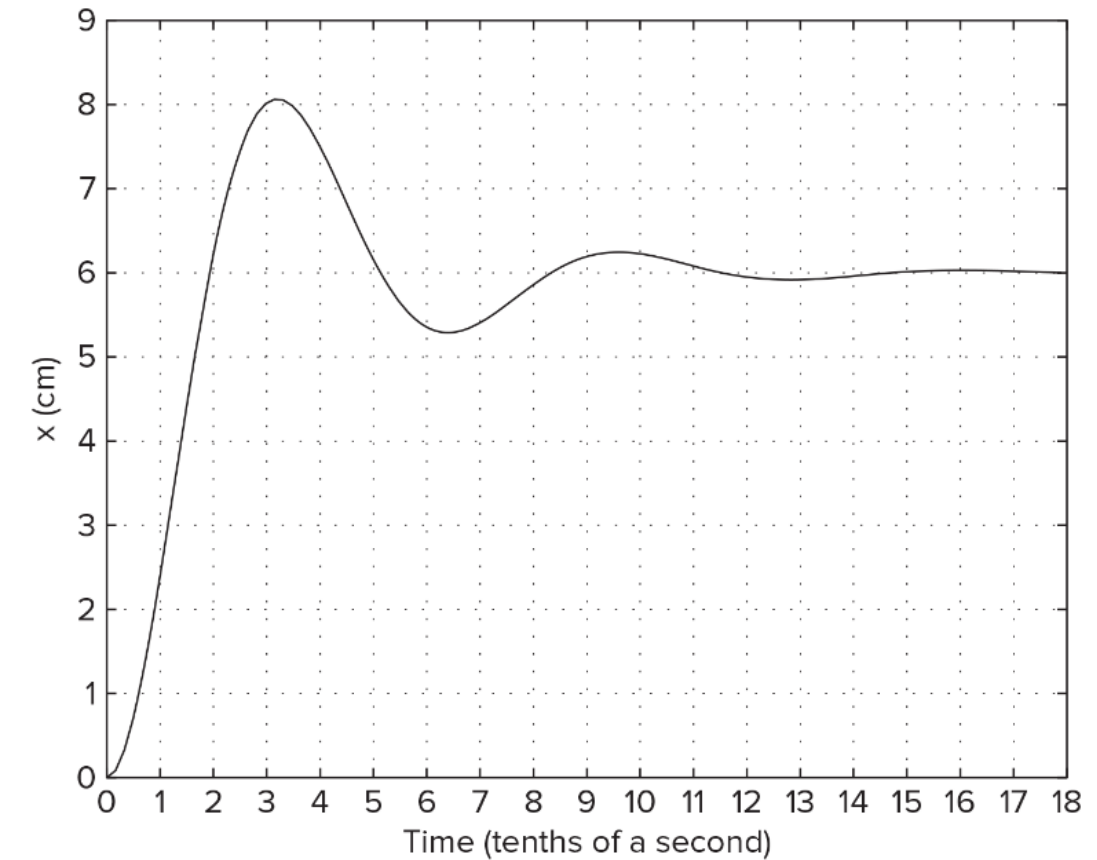
$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$

Estimate the values of m , b , and k .

From the equation of motion, the **transfer function model** of the system is obtained as:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \rightarrow G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

Since the model is a **second-order** transfer function, we can find the model parameters in terms of the **DC-gain**, **damping ratio** and **undamped natural frequency** and determine those values from the step response and estimate the system parameters.



$$G(s) = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\left\{ \begin{array}{l} \omega_n^2 = \frac{k}{m} \\ K\omega_n^2 = \frac{1}{m} \\ 2\zeta\omega_n = \frac{b}{m} \end{array} \right.$$



$$k = \frac{1}{K}$$

$$m = \frac{1}{K\omega_n^2}$$

$$b = \frac{2\zeta}{K\omega_n}$$

System Modeling via Step Response

Example 1

Figure shows the response of a mass-spring-damper system to a step input of magnitude $6 \times 10^3 \text{ N}$.

The equation of motion is:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$

Estimate the values of m , b , and k .

From the step response we have **steady-state value**, **peak-time** and **maximum deviation from the steady-state value**:

$$x_{ss} = 0.06 \text{ m}, \quad t_p = 0.32 \text{ sec}, \quad M_p = 0.081 - 0.06 = 0.021 \text{ m}$$

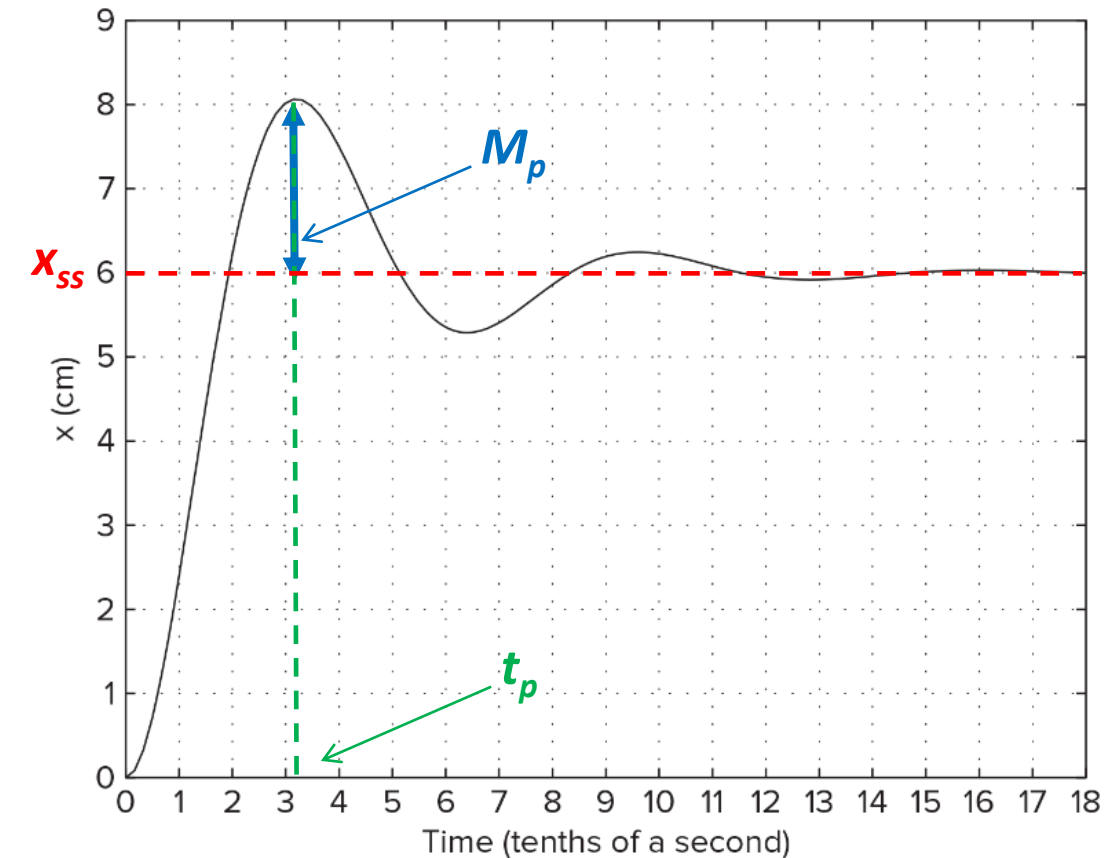
The **DC-gain** of system is obtained as:

$$K = \frac{\Delta x}{\Delta f} = \frac{x_{ss} - x_0}{f_{ss} - f_0} = \frac{0.06 - 0}{6000 - 0} = 10^{-5} \text{ m/N} \rightarrow \boxed{K = 10^{-5} \text{ m/N}}$$

The **damping ratio** is determined as below

$$O.S. = \frac{M_p}{x_{ss}} = \frac{0.021}{0.06} = 0.35$$

$$\zeta = \frac{-\ln(O.S.)}{\sqrt{\pi^2 + \ln^2(O.S.)}} = \frac{-\ln(0.35)}{\sqrt{\pi^2 + \ln^2(0.35)}} \rightarrow \boxed{\zeta = 0.32}$$



System Modeling via Step Response

Example 1

Figure shows the response of a mass-spring-damper system to a step input of magnitude $6 \times 10^3 \text{ N}$.

The equation of motion is:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$

Estimate the values of m , b , and k .

The undamped natural frequency is calculated as below

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \rightarrow 0.32 = \frac{\pi}{\omega_n \sqrt{1 - (0.32)^2}} \rightarrow \boxed{\omega_n = 10.36 \text{ rad/s}}$$

The system parameters are estimated as:

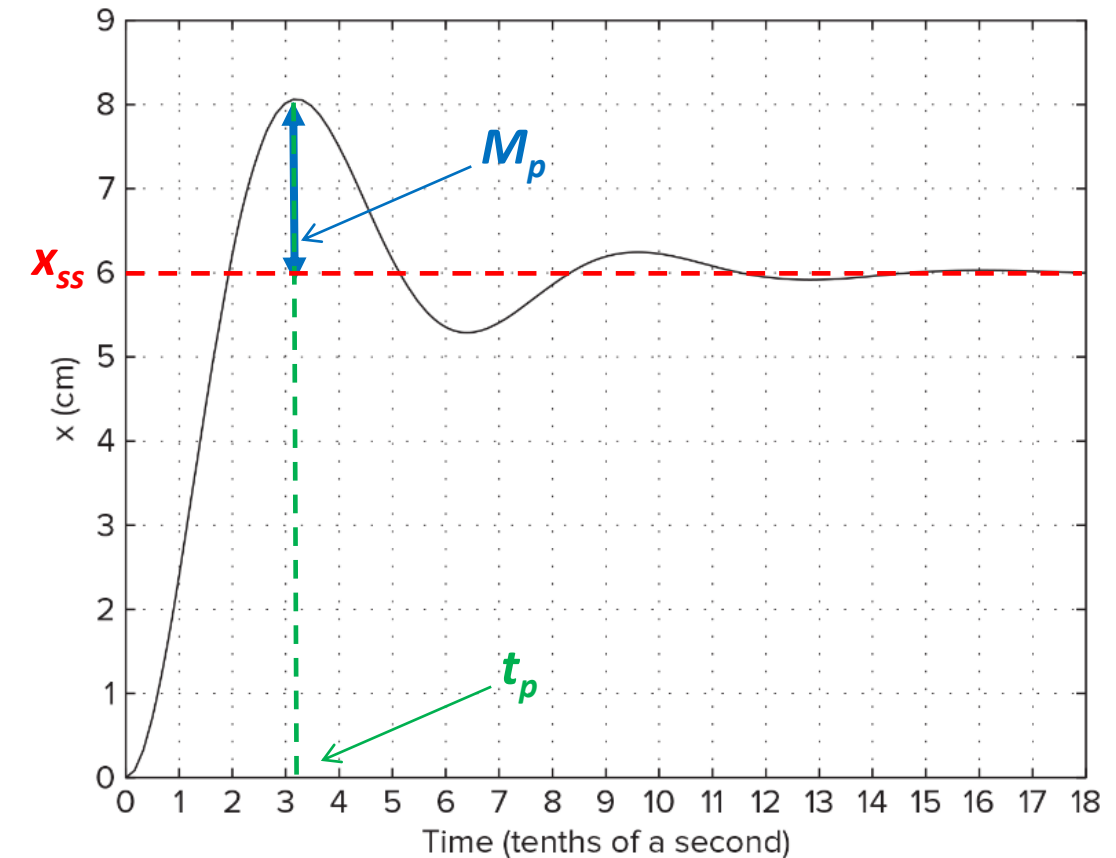
$$k = \frac{1}{K} \rightarrow k = \frac{1}{10^{-5}} = 10^5 \text{ N/m}$$

$$m = \frac{1}{K\omega_n^2} \rightarrow m = \frac{1}{10^{-5}(10.36)^2} = 932 \text{ kg}$$

$$b = \frac{2\zeta}{K\omega_n} \rightarrow b = \frac{2(0.32)}{10^{-5}(10.36)} = 6178 \text{ N.s/m}$$

The transfer function model is:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{932s^2 + 6178s + 10^5}$$

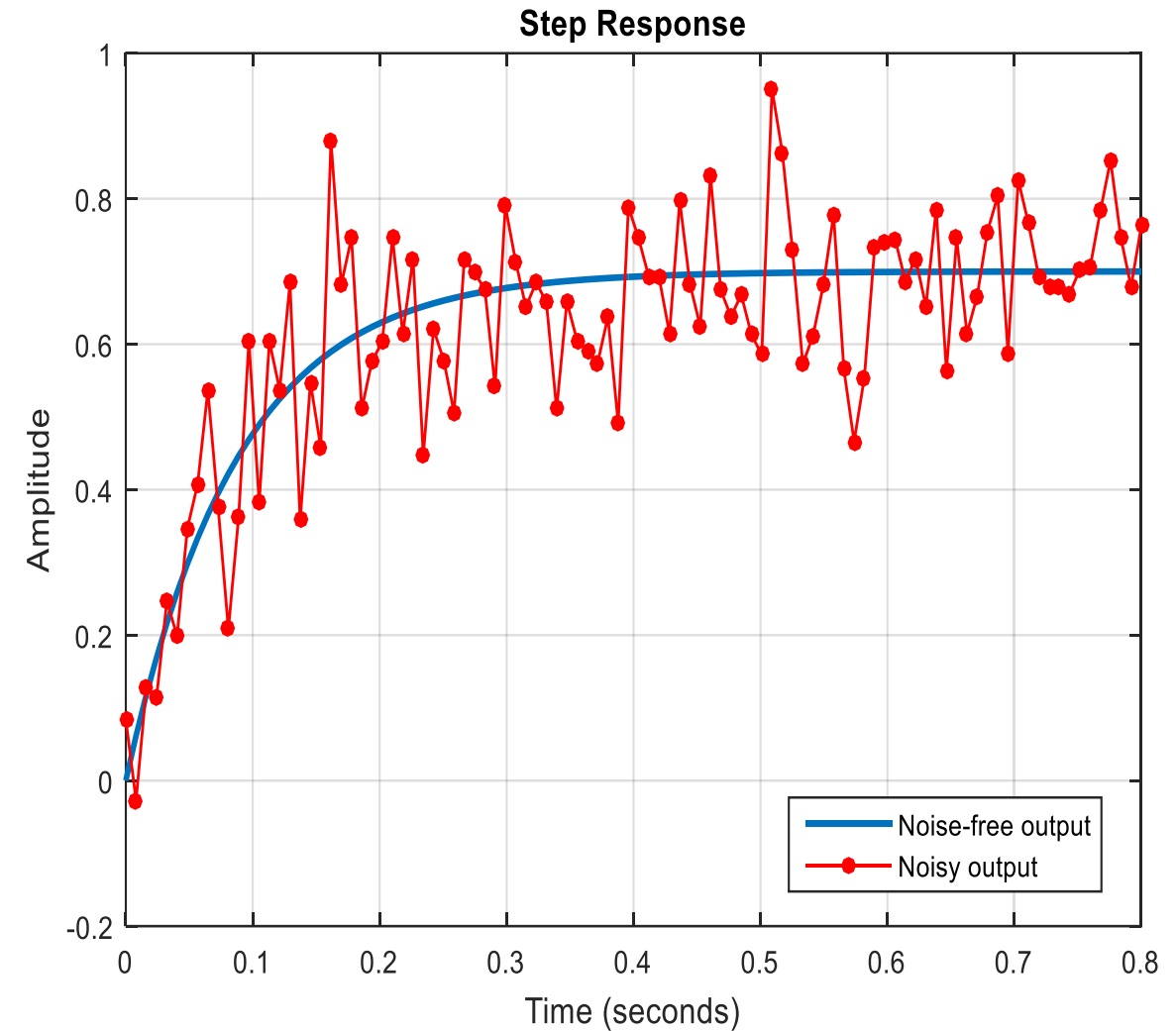


System Modeling via Step Response

❑ Disadvantages of Transient Response Modeling

- Transient-response analysis is **very sensitive to noise**.
- If measurement of the output signal **contaminated by a considerable noise level**, it will be hard to assess the system properties by a single measurement.

We can estimate a transfer function model using the **input/output data** and applying **System Identification methods**.



$$G(s) = ?$$

Modern System Identification

- Three modeling approaches are common in the field of system modeling and identification:
 - **White-Box Modeling:**
 - The system is entirely known.
 - The model order, model parameters and model structure are known.
 - The system can be modeled by differential equations derived from first-principles.
 - **Grey-Box Modeling:**
 - The system is not entirely known.
 - A certain model is constructed based on both insight into the system and experimental data.
 - The model still have some unknown parameters to be estimated using system identification.
 - **Black-Box Modeling:**
 - No prior model is available. Most of the system identification algorithms are of this type.
- The field of **Modern System Identification** uses **statistical methods** to build mathematical models of dynamical systems from measured experimental data.
- System identification also includes the optimal design of experiments for efficiently generating informative data for fitting such models as well as model reduction.
- A much more common approach is therefore to start from measurements of the behavior of the system (output response) and the external influences (inputs to the system) and try to determine a mathematical relation between them without going into the details of what is actually happening inside the system.
- **This approach is called System Identification.**

System Identification Procedure

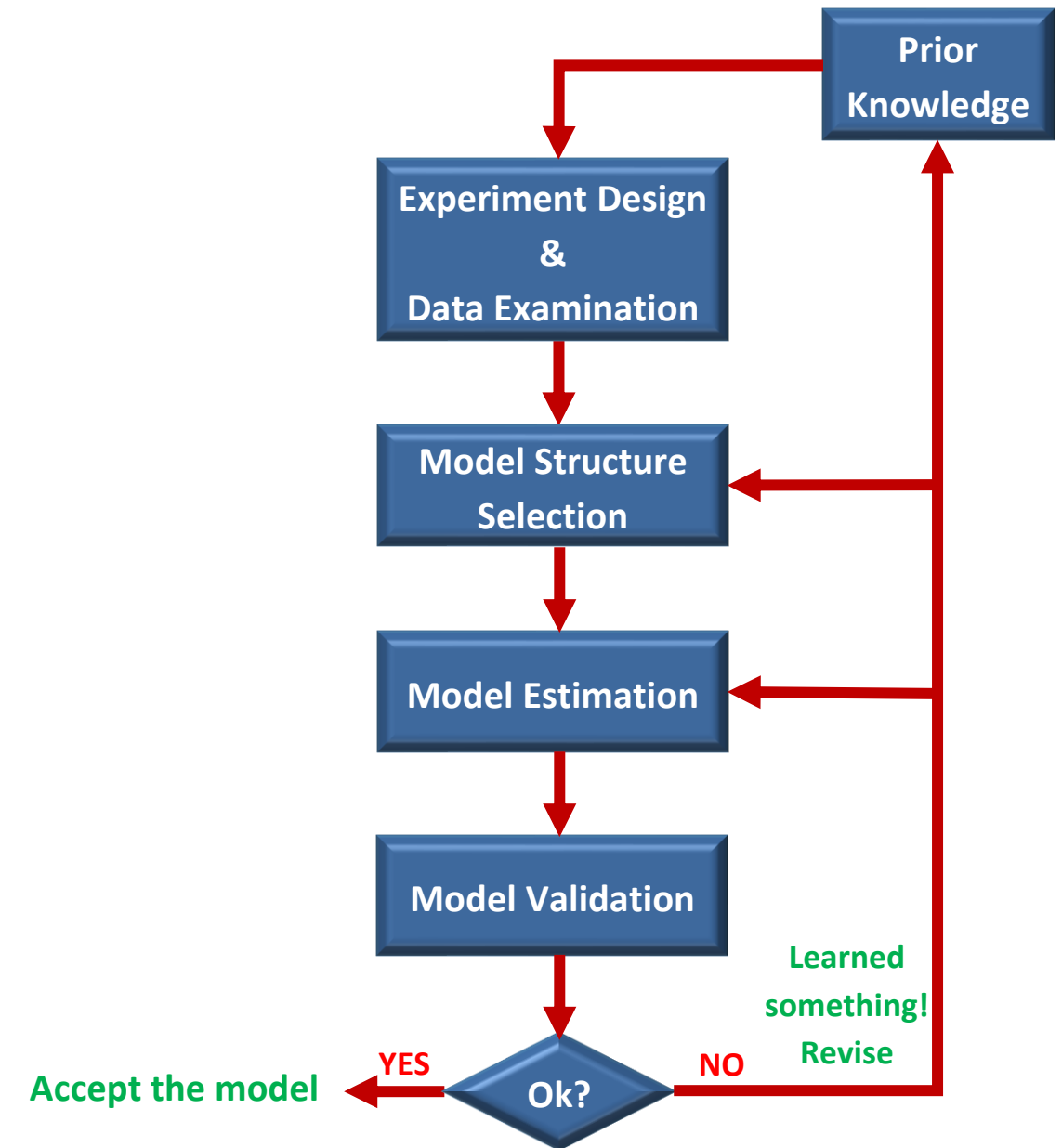
- **System Identification** is an **iterative** procedure, and it is often necessary to go back and repeat earlier steps.

■ Prior Knowledge

- Purpose of Modeling
 - Control System Design
- Grey-box Identification
 - Some part of the system is known
 - Model Order, Dominant Pole Locations, An Integrator,
- Black-box Identification
 - No prior knowledge about the system

■ Experiment Design & Data Examination

- Choice of Input Signal and I/O Data Collection
- I/O Data Examination
 - Aliasing, Outliers and Trends, Noise Filtering
- Preliminary Diagnostic Experiments
 - Frequency Response Analysis
 - Bode Diagrams
 - Time Response Analysis
 - Impulse Response
 - Step Response

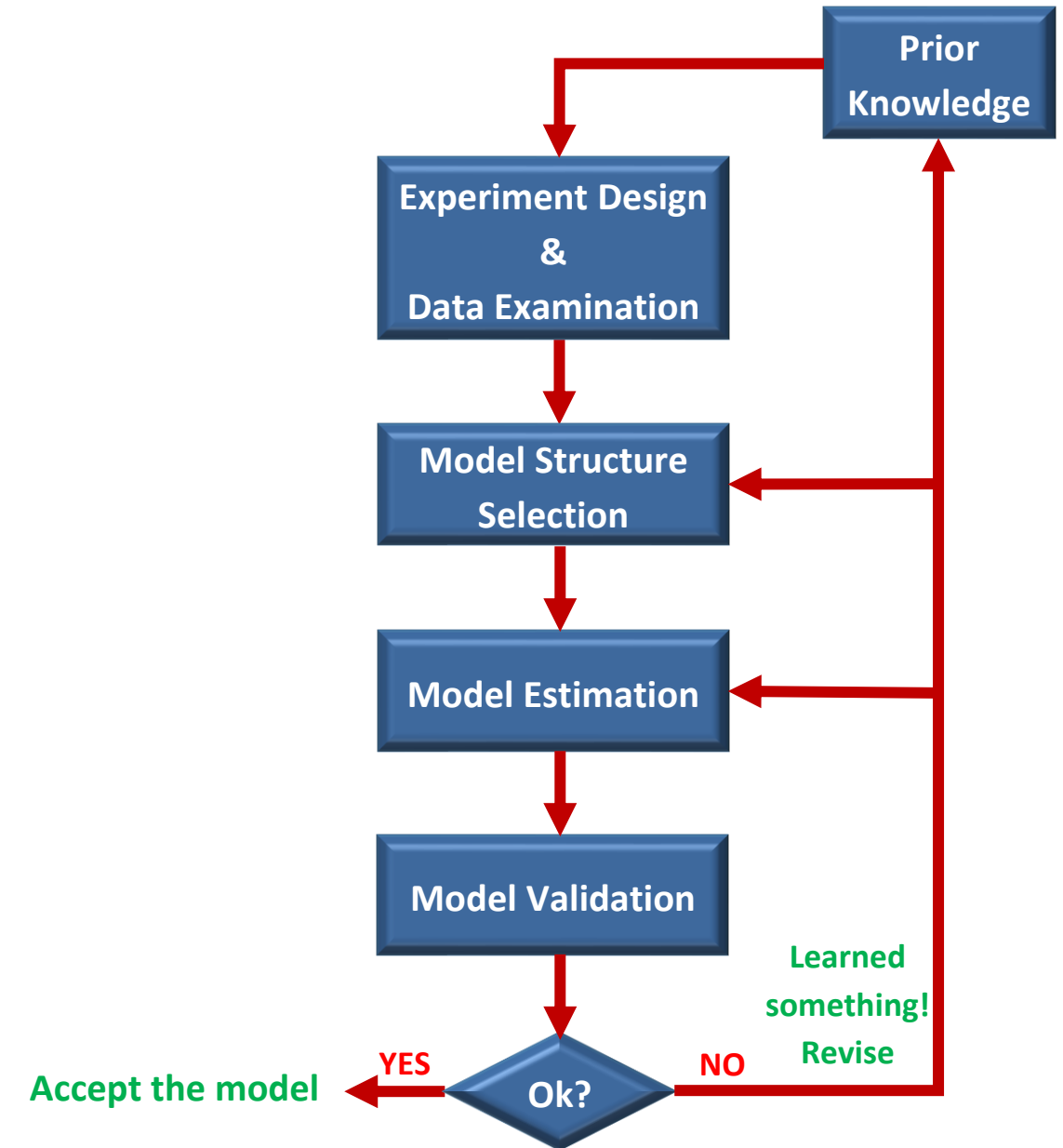


System Identification Procedure

- **System Identification** is an iterative procedure, and it is often necessary to go back and repeat earlier steps.

- **Parametric Model Structure Selection**

- Continuous Time Models
 - Transfer Function Model
 - State Space Model
- Discrete Time Models
 - Linear Regression Models
 - Transfer Function Models
 - Box-Jenkins (BJ) Model
 - Output-Error (OE) Model
 - ARX Model
 - ARMAX Model
 - State Space Models
 - Time Series Models
 - AR Model
 - MA Model
 - ARMA Model



System Identification Procedure

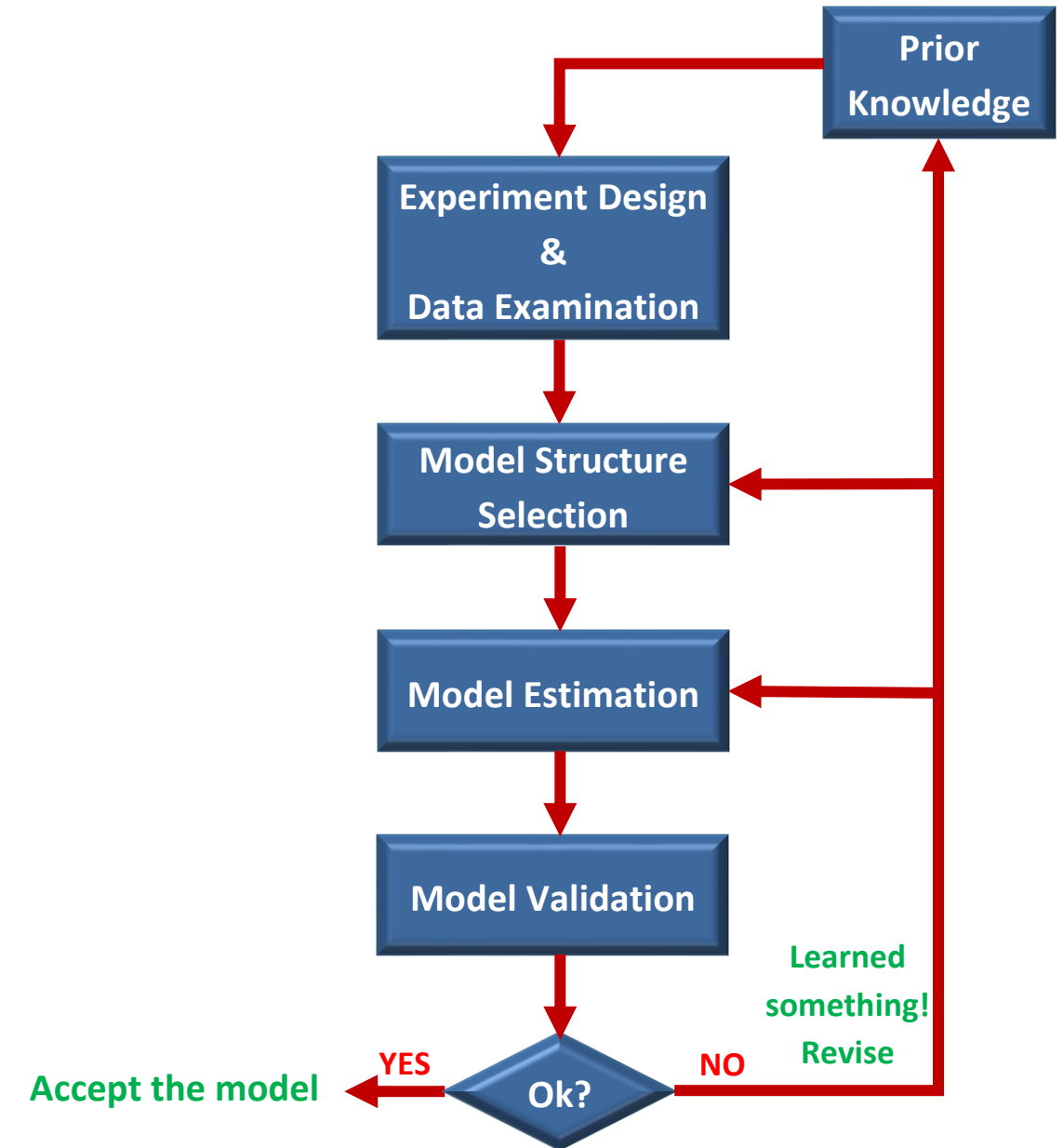
- **System Identification** is an **iterative** procedure, and it is often necessary to go back and repeat earlier steps.

■ Model Estimation Techniques

- Nonparametric Methods
 - Spectral Analysis
 - Correlation Analysis
- Parametric Methods
 - Least Squares Method

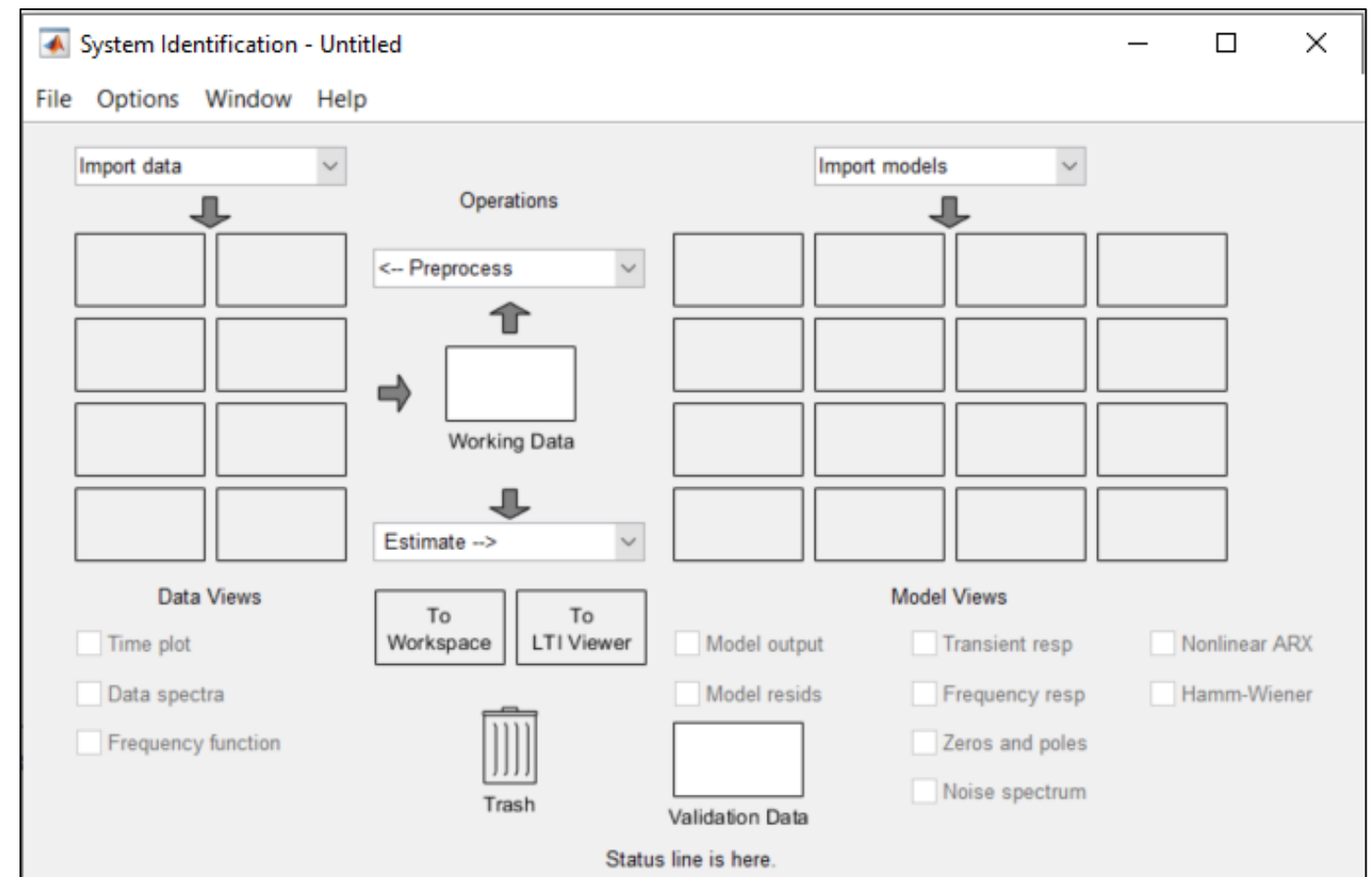
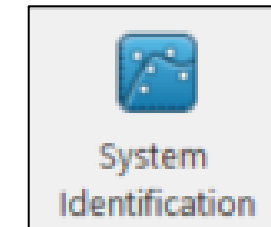
■ Model Validation Techniques

- Simulation
- Cross-Validation
- Model Validity Criterion
 - Mean-Squares Error (MSE)
 - Akaike's Final Prediction Error (FPE)
- Pole-Zero Plots
- Bode Diagram
- Residual Analysis
 - Auto-correlation Analysis
 - Cross-correlation Analysis



Identification of Transfer Function Models

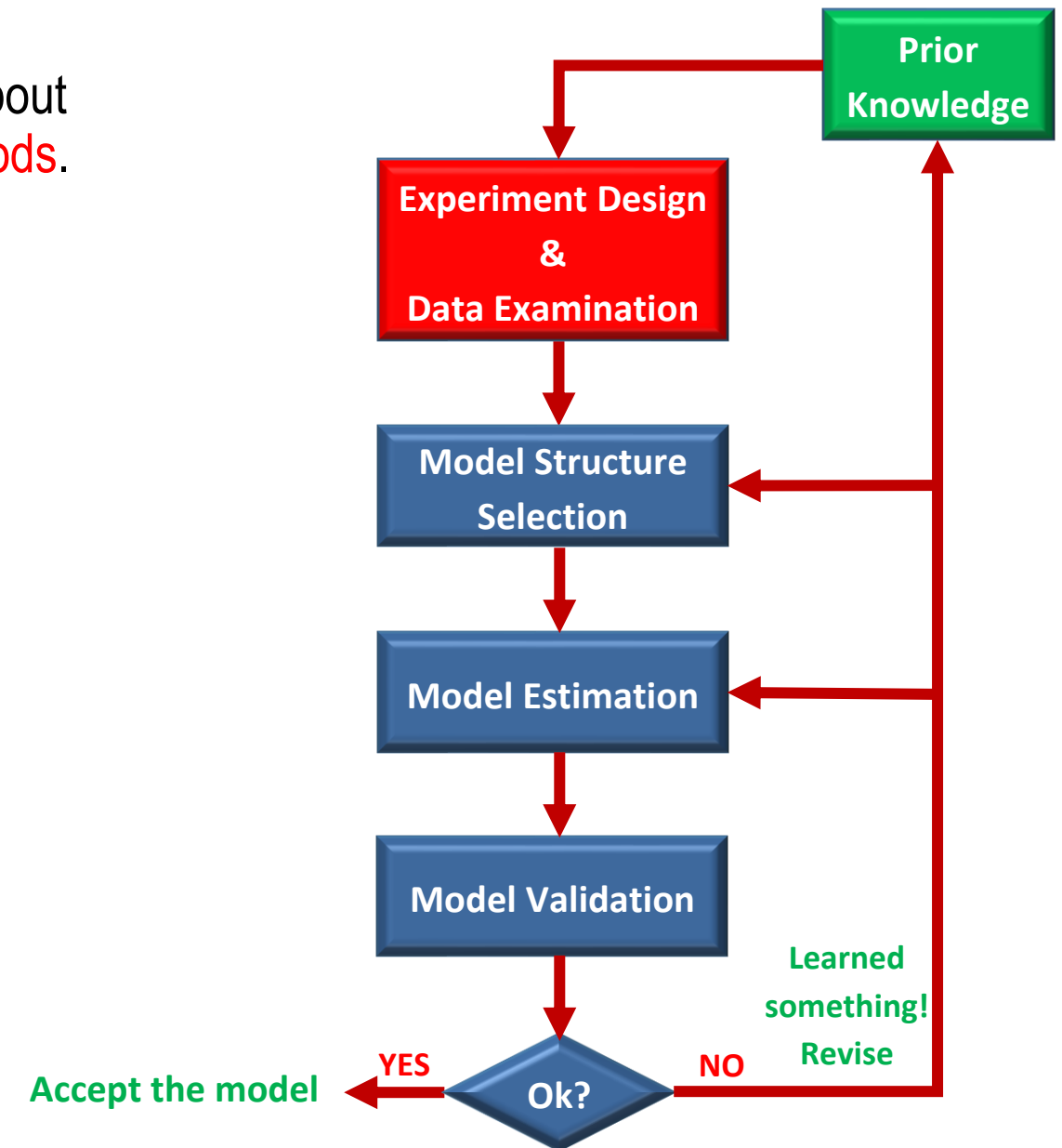
- **MATLAB System Identification Toolbox** provides [line commands](#) and an [app](#) for constructing mathematical models of dynamic systems from measured input-output data.
- It enables us to create and use models of dynamic systems that are not easily be modeled from first principles or specifications.
- We can use [time-domain](#) and [frequency-domain](#) input-output data to identify [continuous-time](#) and [discrete-time transfer functions](#), [process models](#), and [state-space](#) models.
- [System Identification Toolbox](#) enables us to create models from measured input-output data.
 - Analyze and process the input-output data
 - Select suitable model structure and order
 - Estimate model parameters
 - Validate the model accuracy



System Identification Procedure

❑ Experiment Design & Data Examination

- The **goal** is to **collect the I/O data** and to obtain some **primary knowledge** about important characteristics of system using **non-parametric models** and **methods**.
- **Choice of Input Signal and I/O Data Collection**
 - Pseudo Random Binary Sequence (PRBS)
- **I/O Data Examination**
 - Aliasing, Outliers and Trends
 - Noise Filtering
- **Preliminary Diagnostic Experiments**
 - **Frequency-Domain Analysis**
 - Non-parametric Models in Frequency-domain
 - Bode Plot
 - **Time-Domain Analysis**
 - Non-parametric Models in Time-domain
 - Impulse Response
 - Step Response

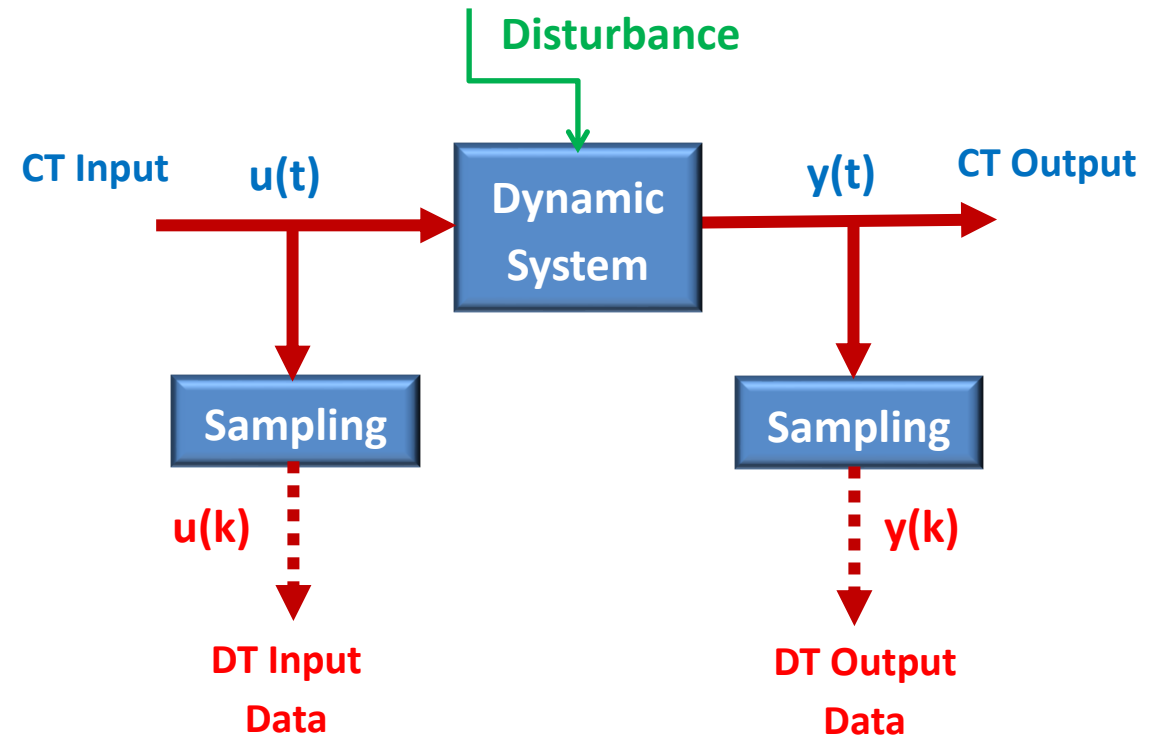


Choice of Input Signal and Data Collection

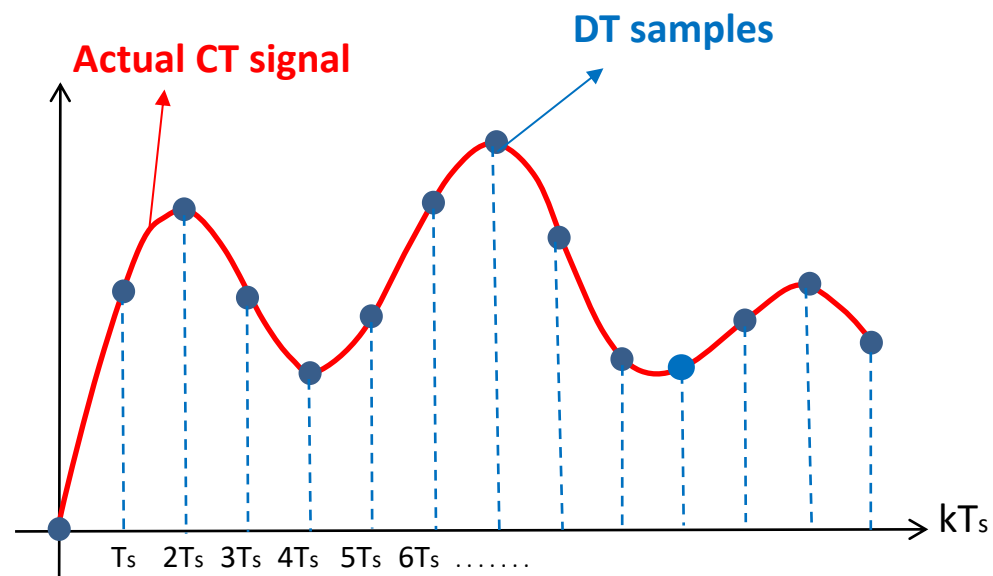
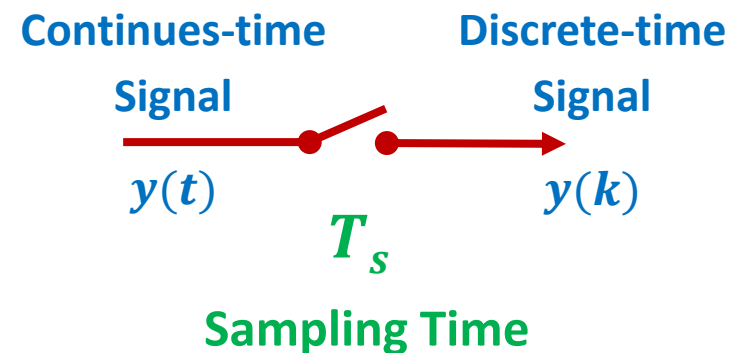
- In system identification, models are built from **experimental data**.
- Such data is obtained by **exciting the system** with an input and observing its response at **regular intervals**.
- This means we are dealing with **samples of discrete-time data**.

□ Sampling

- **Sampling** is the process of converting the **continues-time** signal to a **discrete-time** signal



$$\{u(k), y(k) | k = 1, 2, \dots, N\}$$



Choice of Input Signal and Data Collection

□ Sampling Rate

- Selecting the **correct sampling rate** is essential, when we convert a continuous-time signal to a discrete-time signal.

- Sampling Theorem** → The **sampling rate** f_s must be at least **twice faster** than the **highest frequency** contained in the continuous-time signal.

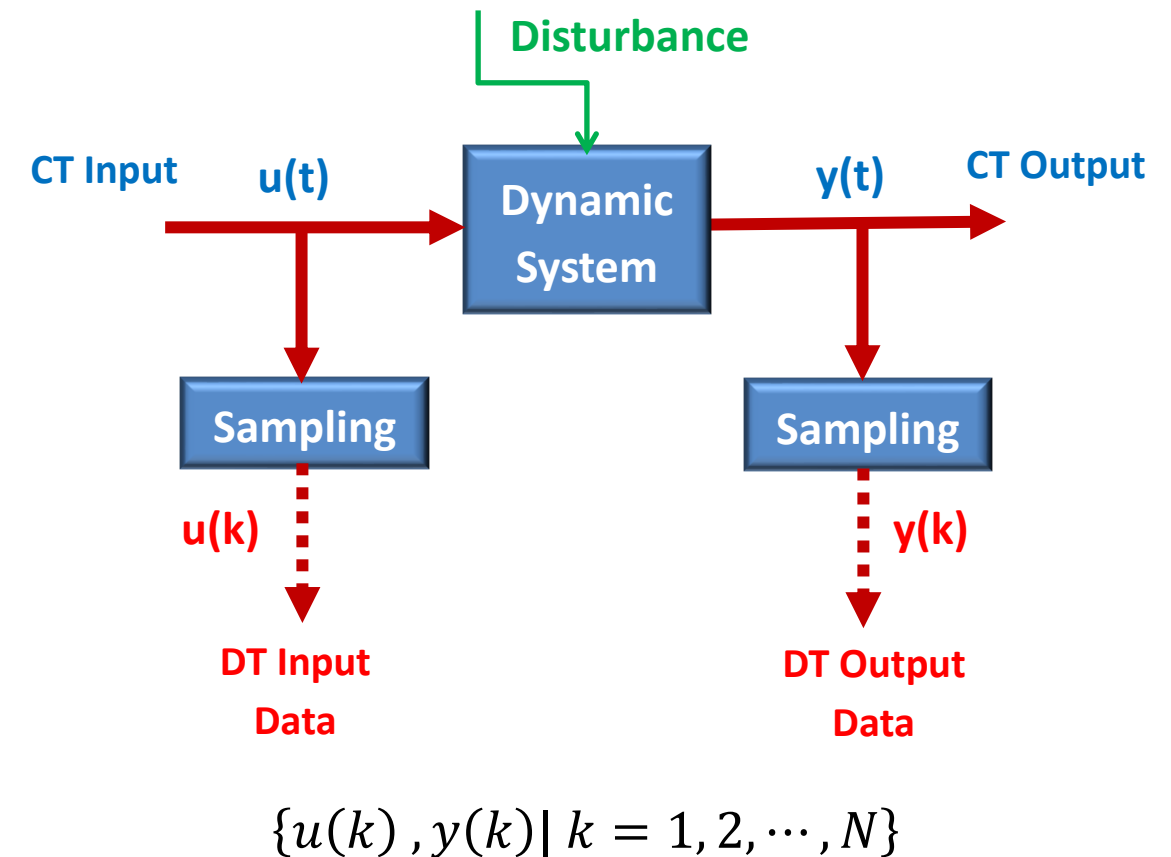
$$f_s \geq 2f_{max}$$

- The **minimum required sampling rate** is called **Nyquist Rate**:

$$\text{Nyquist Rate} = 2f_{max}$$

□ Aliasing Effect

- If the selected sampling rate does not satisfy the **Sampling Theorem condition**, then reconstructed signal from its sampled version leads to a **distortion**. This effect is called **frequency-folding** or **aliasing**.
- If there are **aliasing effects** in data, the **sampling rate** should be increased by considering the **Sampling Theorem condition**.



Choice of Input Signal and Data Collection

Example 2

Consider the continuous-time signal is a sinusoid with the frequency of

$$f = 60\text{Hz}$$

From the **Sampling Theorem** the sampling rate must be selected as

$$f_s \geq 120\text{Hz}$$

Here, the selected sampling frequency is

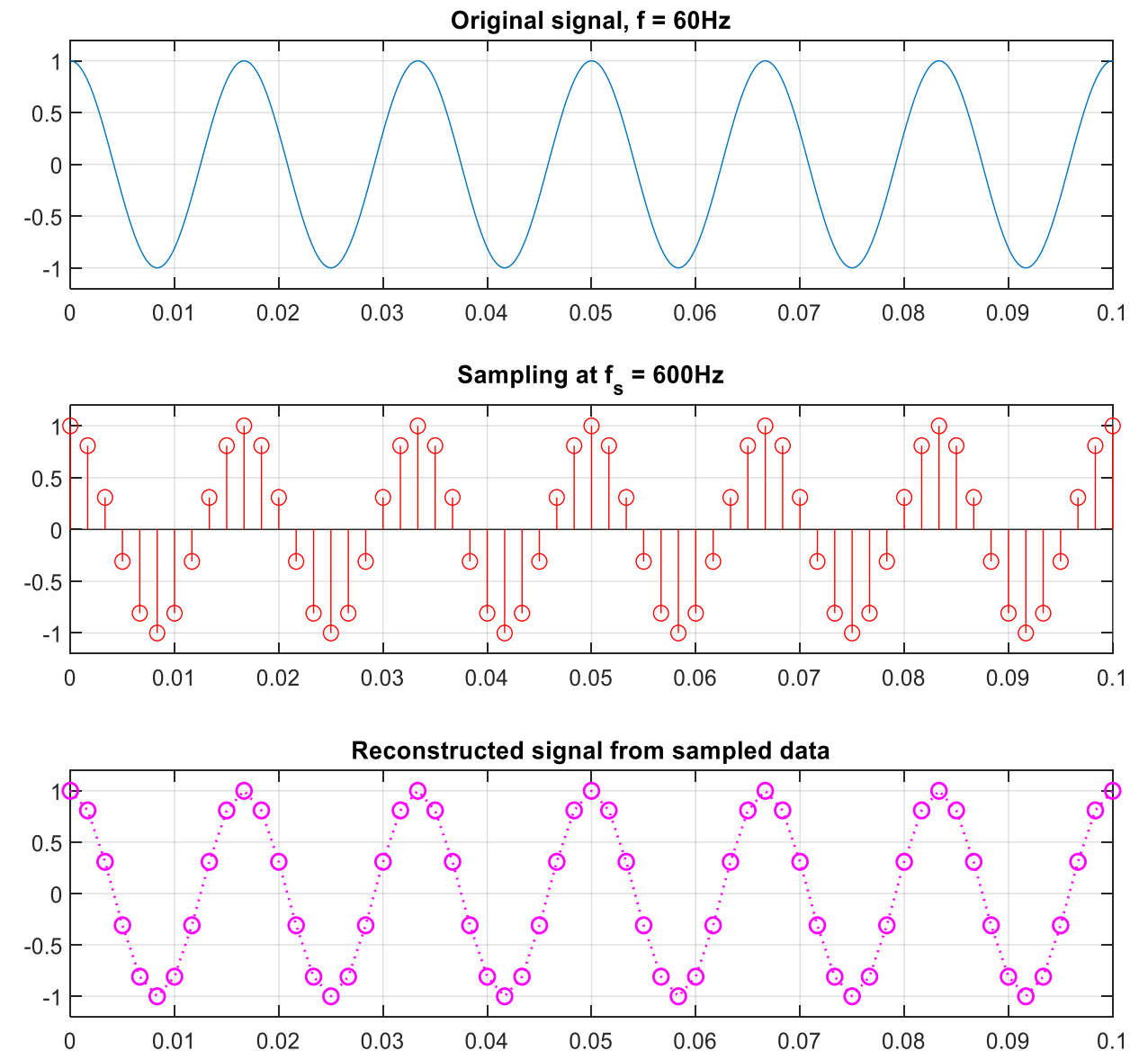
$$f_s = 600\text{Hz}$$

Good Sampling

The selected sampling frequency is well above the **Nyquist rate**, 120Hz.

Therefore, the samples can capture the oscillation of the original signal.

Reconstructed signal is same as the original signal.



Choice of Input Signal and Data Collection

Example 2

Consider the continuous-time signal is a sinusoid with the frequency of

$$f = 60\text{Hz}$$

From the Sampling Theorem the sampling rate must be selected as

$$f_s \geq 120\text{Hz}$$

Here, the selected sampling frequency is

$$f_s = 70\text{Hz}$$

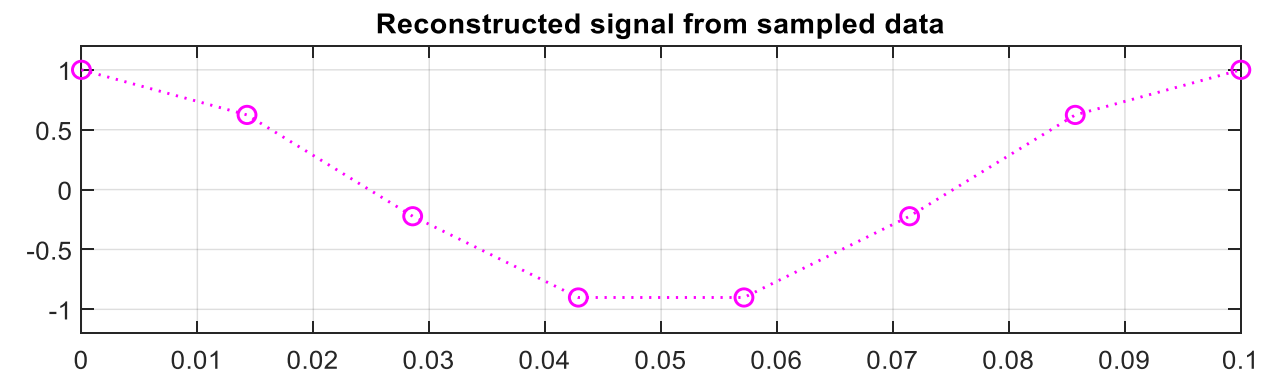
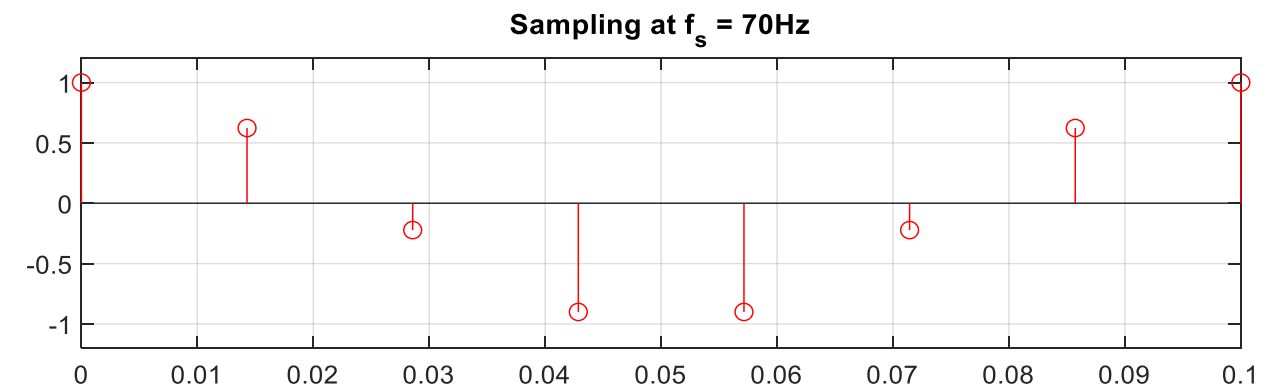
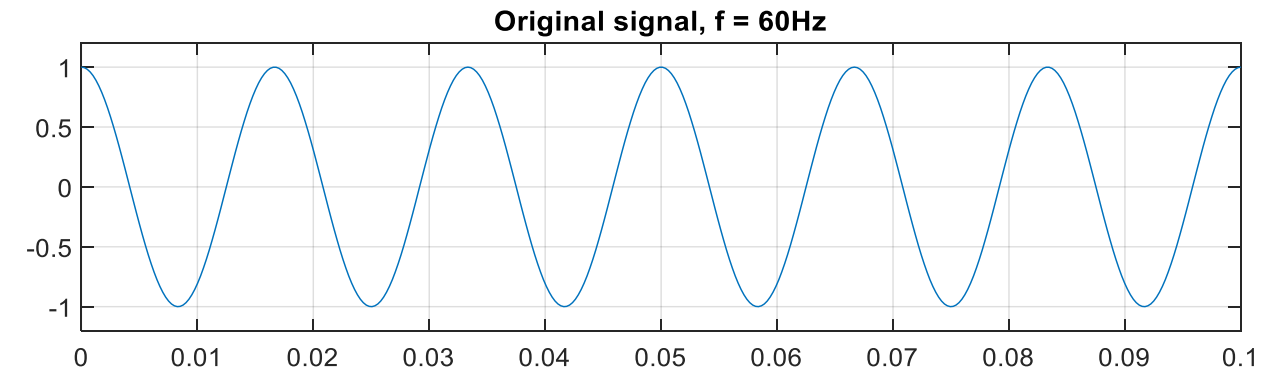


Aliasing

The selected sampling frequency is **too low**.

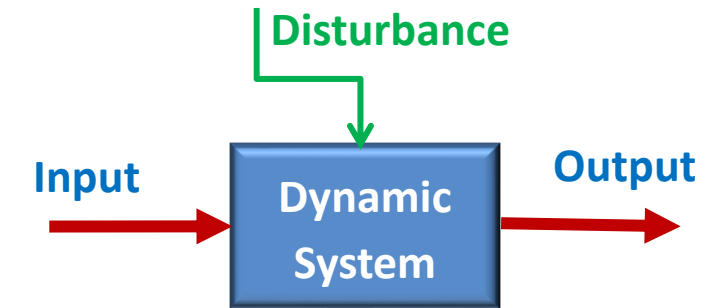
Therefore, the samples cannot capture the oscillation of the original signal, and **aliasing happens**.

The reconstructed signal looked like a **low frequency** sinusoid signal.



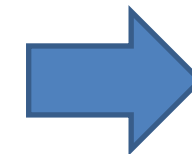
Choice of Input Signal and Data Collection

- The **input signal** of the process plays an important role in system used in a system identification.
- The **input signal** is the only possibility to **influence** the system to **collect information** about its dynamic behavior.



□ The most often used input signals in practice for system identification

- **Step/Impulse Signal** → Time-domain Identification
- **Sinusoid Signal** → Frequency-domain Identification



Non-parametric Methods

- **Pseudo-Random Binary Sequence (PRBS)** → Time-domain Identification
Frequency-domain Identification



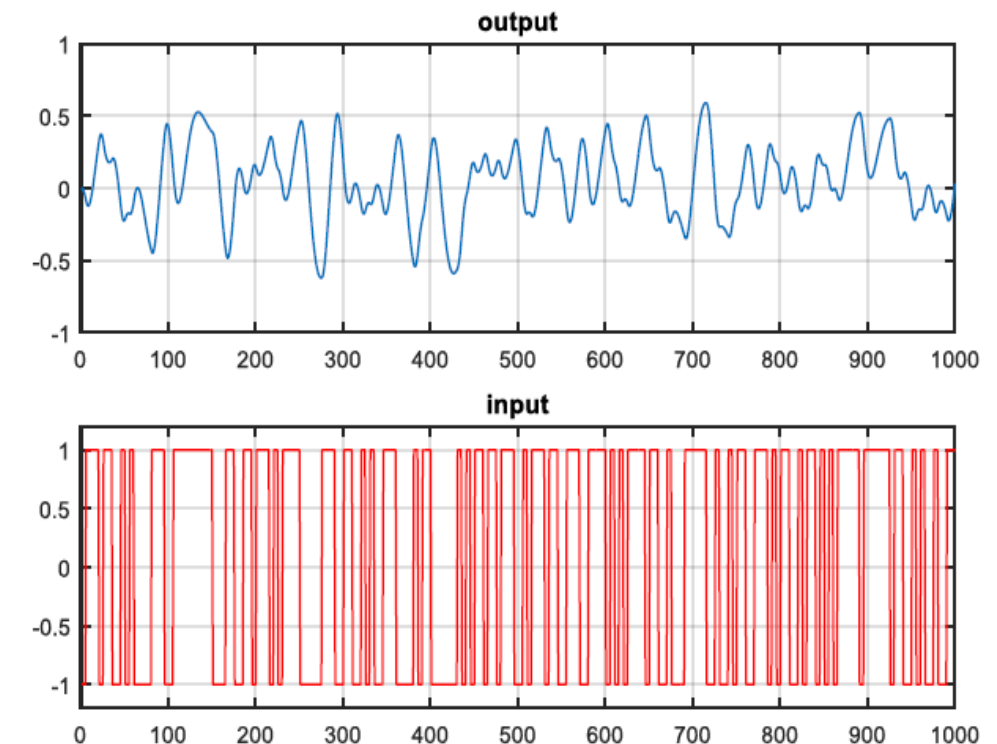
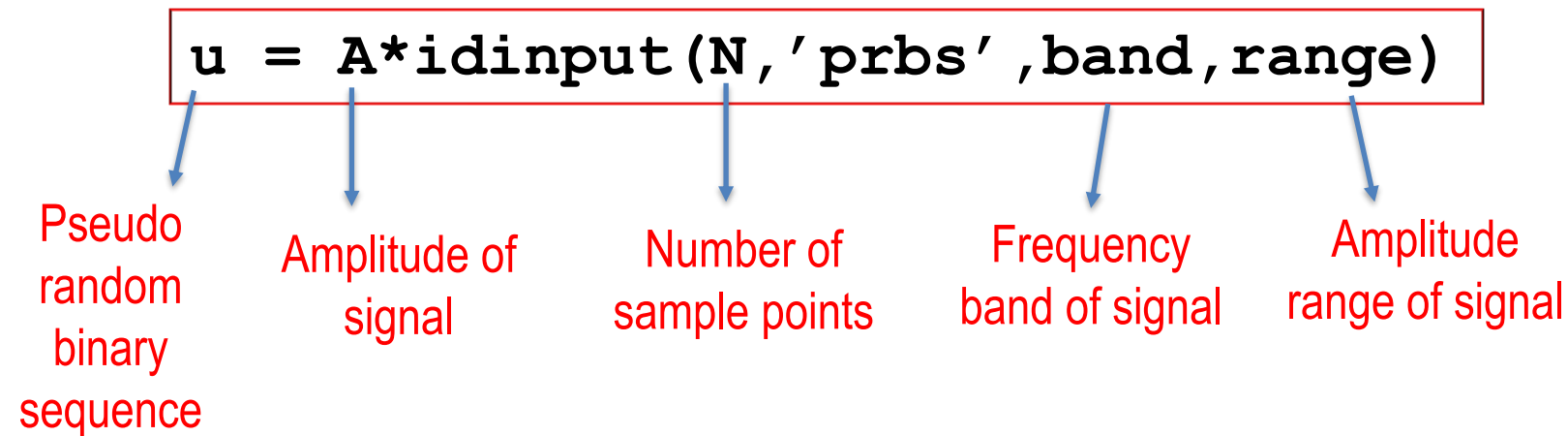
Parametric Methods

In this course, we use
PRBS signal for
system identification

Choice of Input Signal and Data Collection

❑ Pseudo-Random Binary Sequence (PRBS)

- A **Pseudo-random binary sequence (PRBS)** is a common choice of input signal, since it has a **large energy content** in a **large frequency range**.
- **PRBS** is a **square wave (sum of sinusoids)** that randomly changes between +1 and -1
- It is **pseudo random** because we can control when the switch may occur.
- The **idinput** function from **System Identification Toolbox** is available to generate **PRBS** signal.



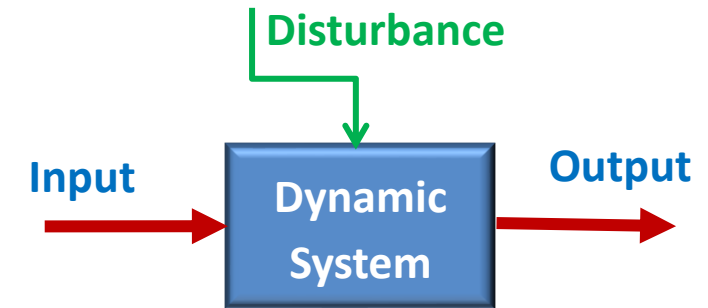
❑ Important Characteristics of Input Signal

- Amplitude
- Frequency Range → Good excitation in the frequency range (**Persistent Excitation**)
- Duration → **Number of Samples**

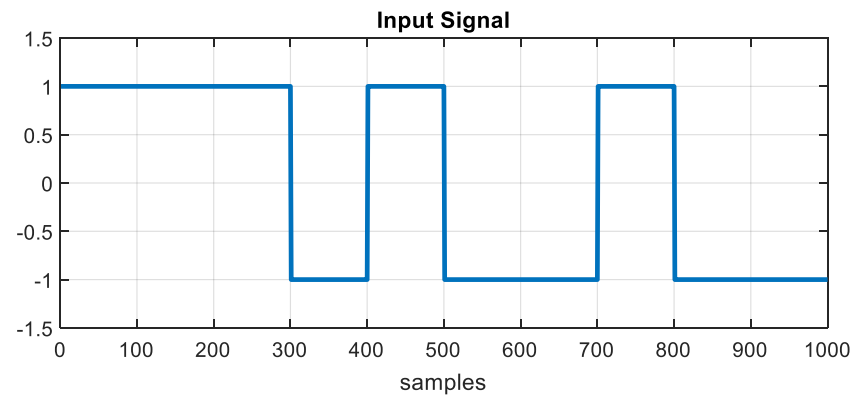
Choice of Input Signal and Data Collection

□ Amplitude

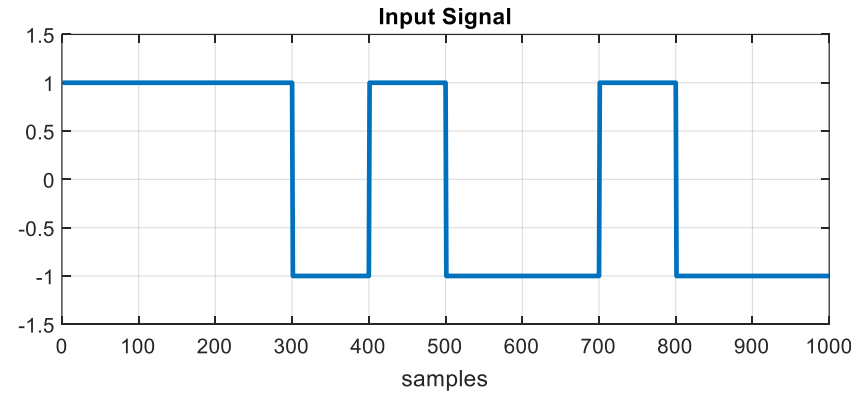
- Amplitude of the input signal should be selected **appropriately**
 - To achieve a **good signal to noise ratio (SNR)**
 - To overcome **friction** and **dead-zone** issues
 - To avoid **saturation** and **non-linearity** range



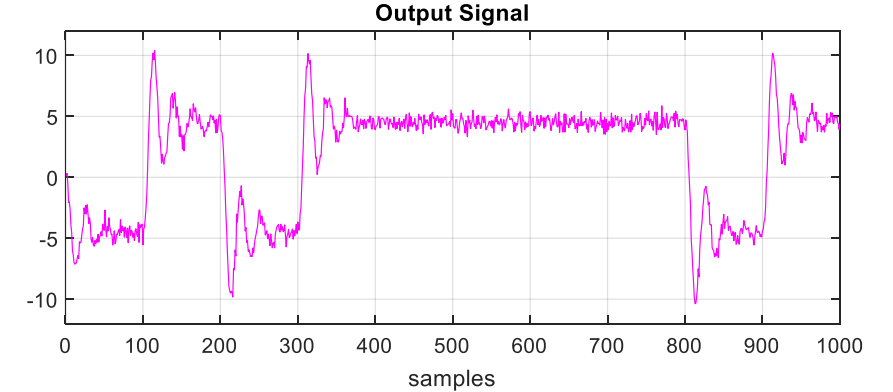
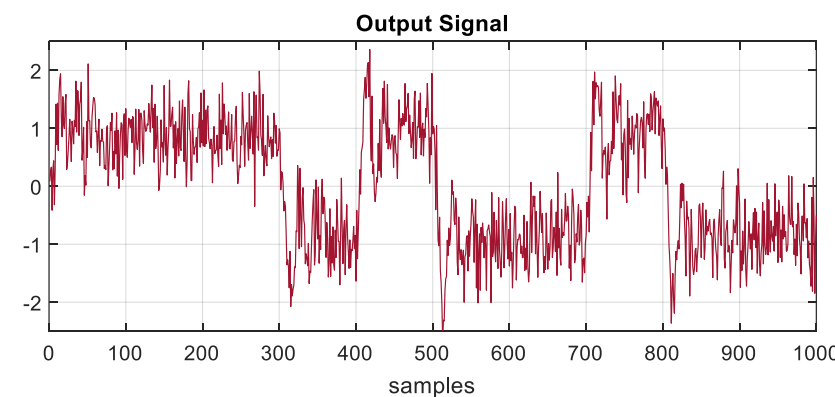
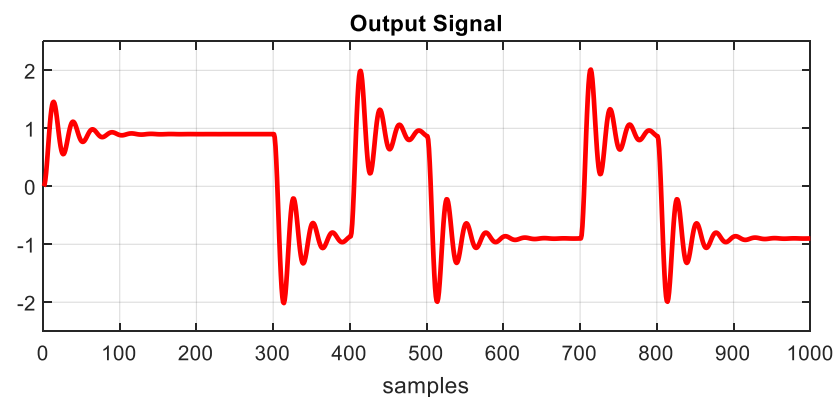
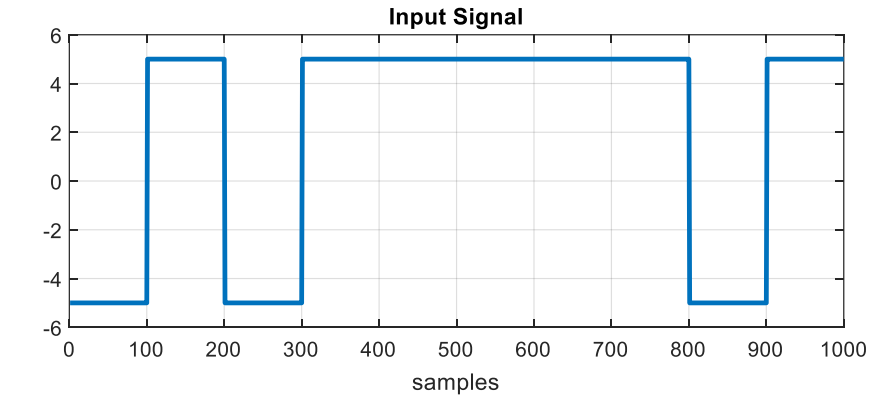
Noise-free System



Low SNR Noisy System



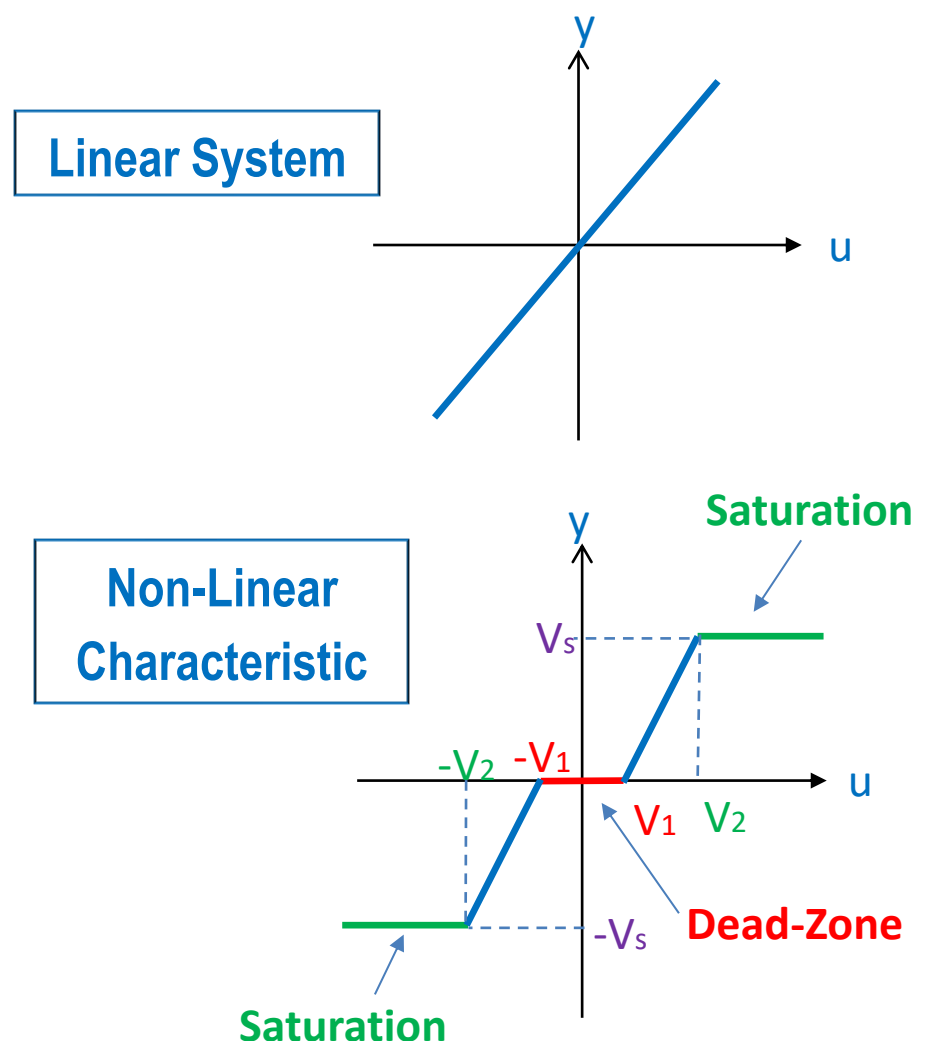
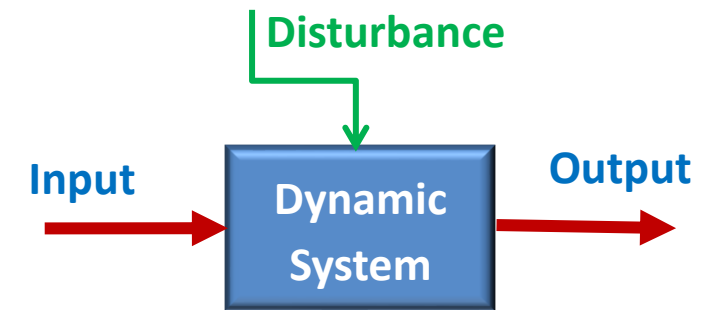
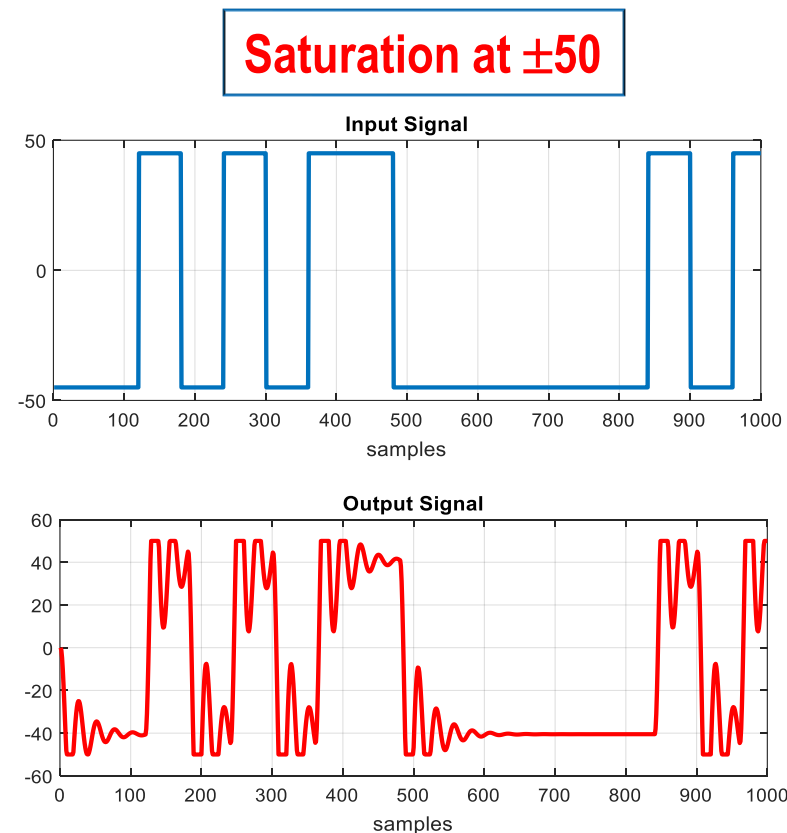
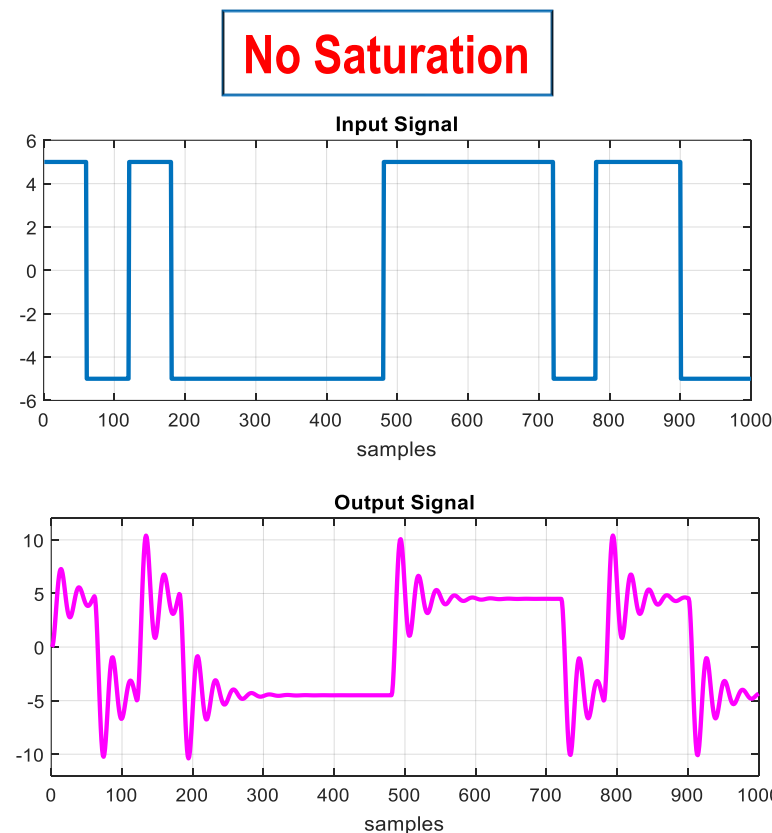
High SNR Noisy System



Choice of Input Signal and Data Collection

□ Amplitude

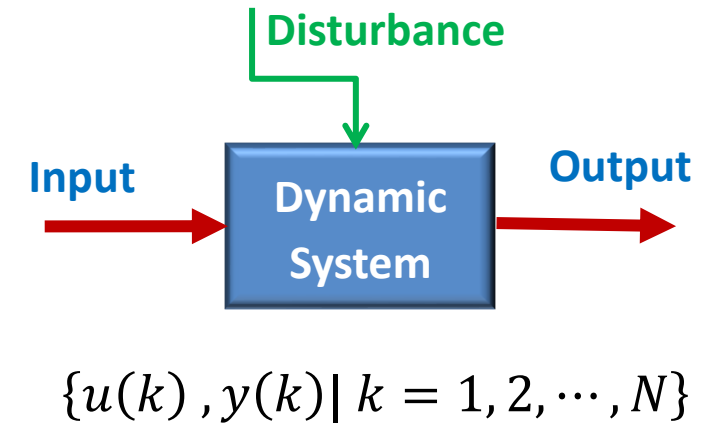
- Amplitude of the input signal should be selected **appropriately**
 - To achieve a **good signal to noise ratio (SNR)**
 - To overcome **friction** and **dead-zone** issues
 - To avoid **saturation** and **non-linearity** range



- The **amplitude** may not be chosen larger than the range in which the **linearity assumption** holds.

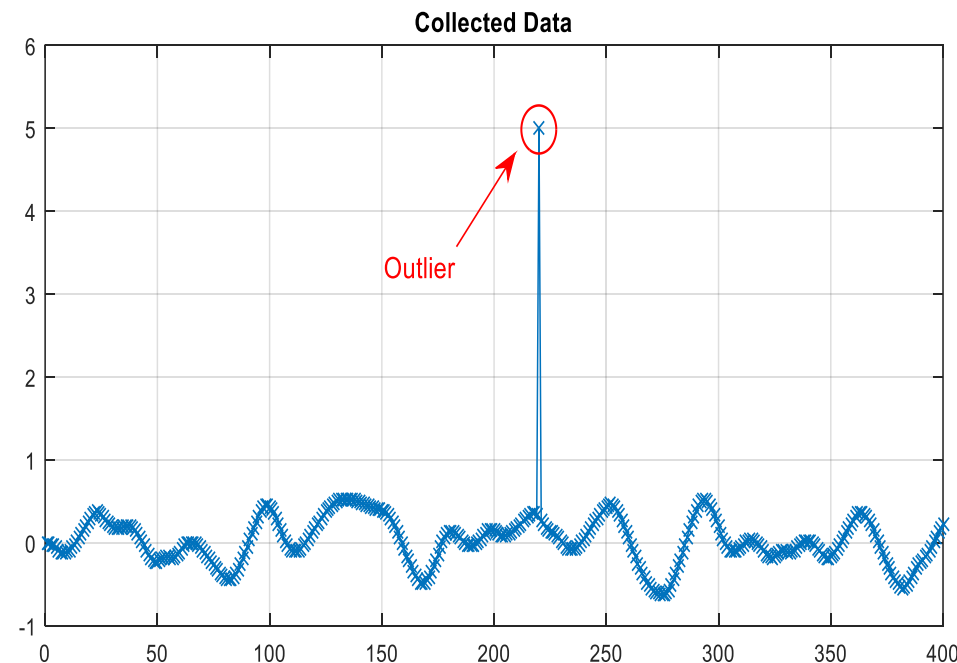
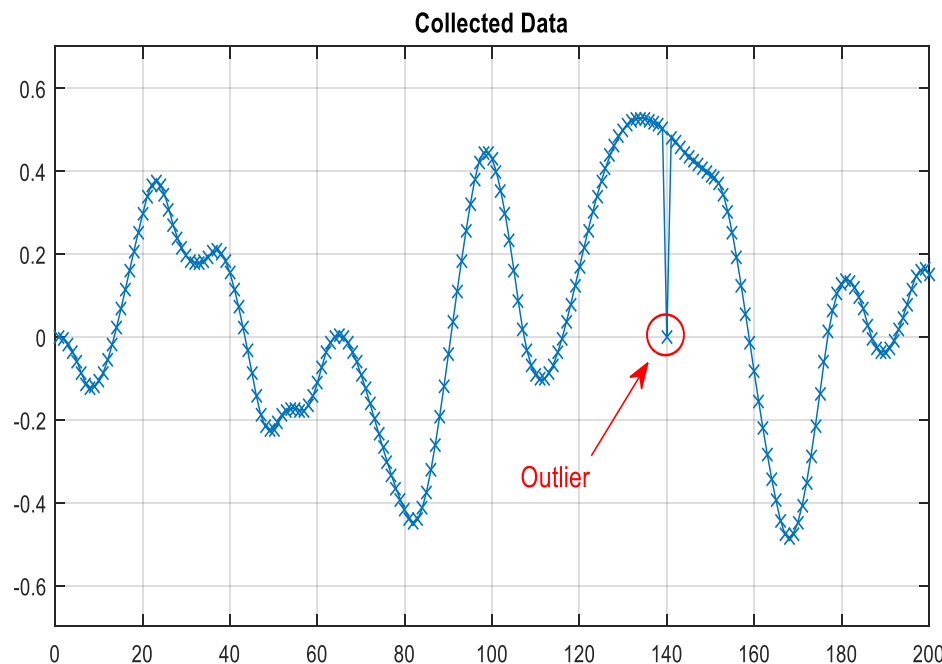
Input-Output Data Examination

- Assume that an experiment has been performed, and we have the input and output sequences.
- Check the data manually via plots and look for:
 - Aliasing Effect
 - Outliers
 - Trends or DC-offset



□ Outliers

- In practice, the **data acquisition equipment** is **not perfect**.
- It may be that certain measured values are in obvious error due to **measurement failures**.
- Such bad values are often called **outliers**.



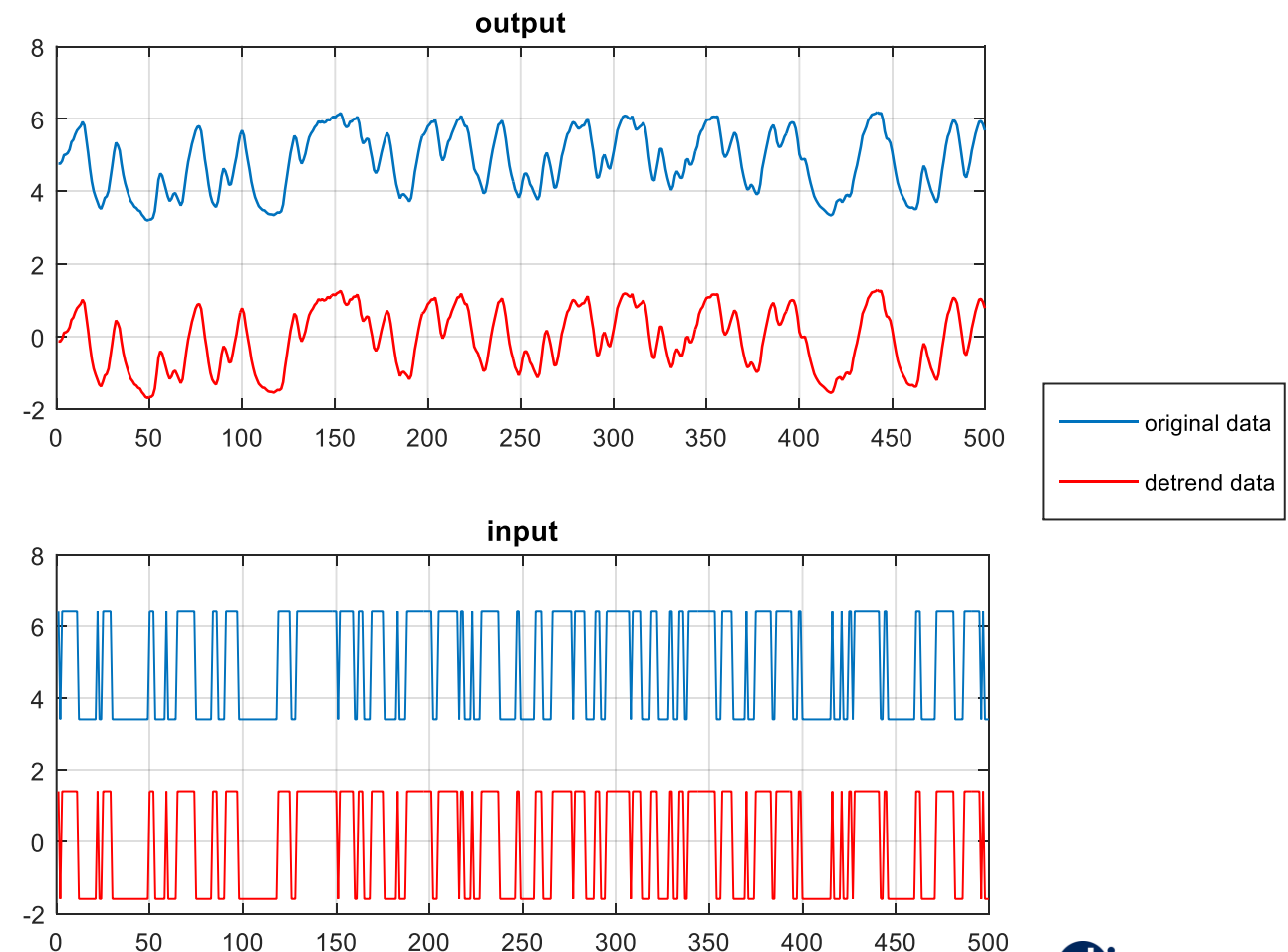
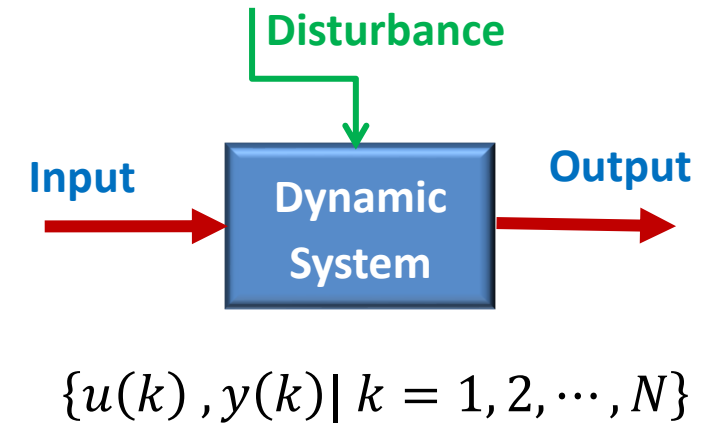
Outliers may have a substantial negative effect on the estimation and should simply be **removed** from the data.

Input-Output Data Examination

❑ Trends or DC-offset

- Measured signals often show **low-frequency drifts** or **non-zero means**, which may have a bad effect on identification results if they are not specifically accounted for.
- Linear trends** in data, such as **low-frequency drifts** or **non-zero means**, should be removed from the I/O data.
- A **linear trend** can be removed by **dtrend** command in MATLAB

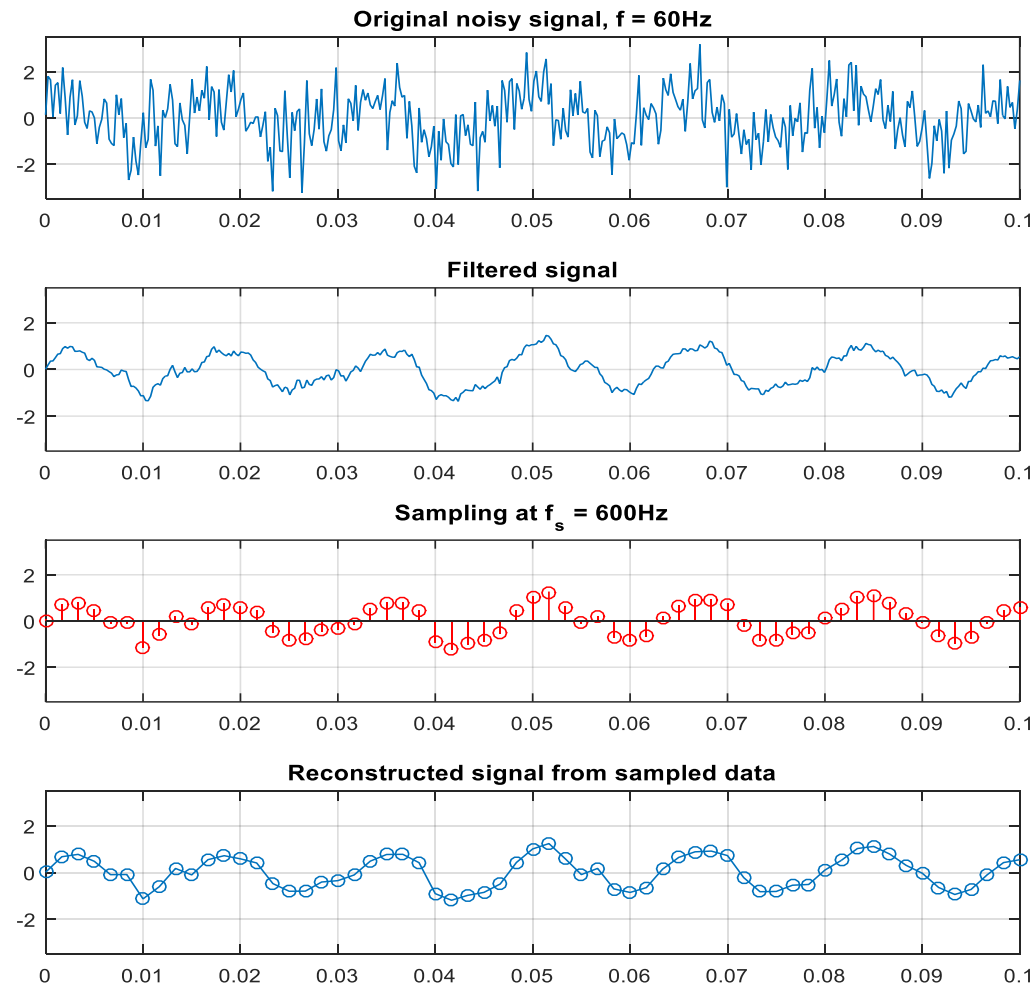
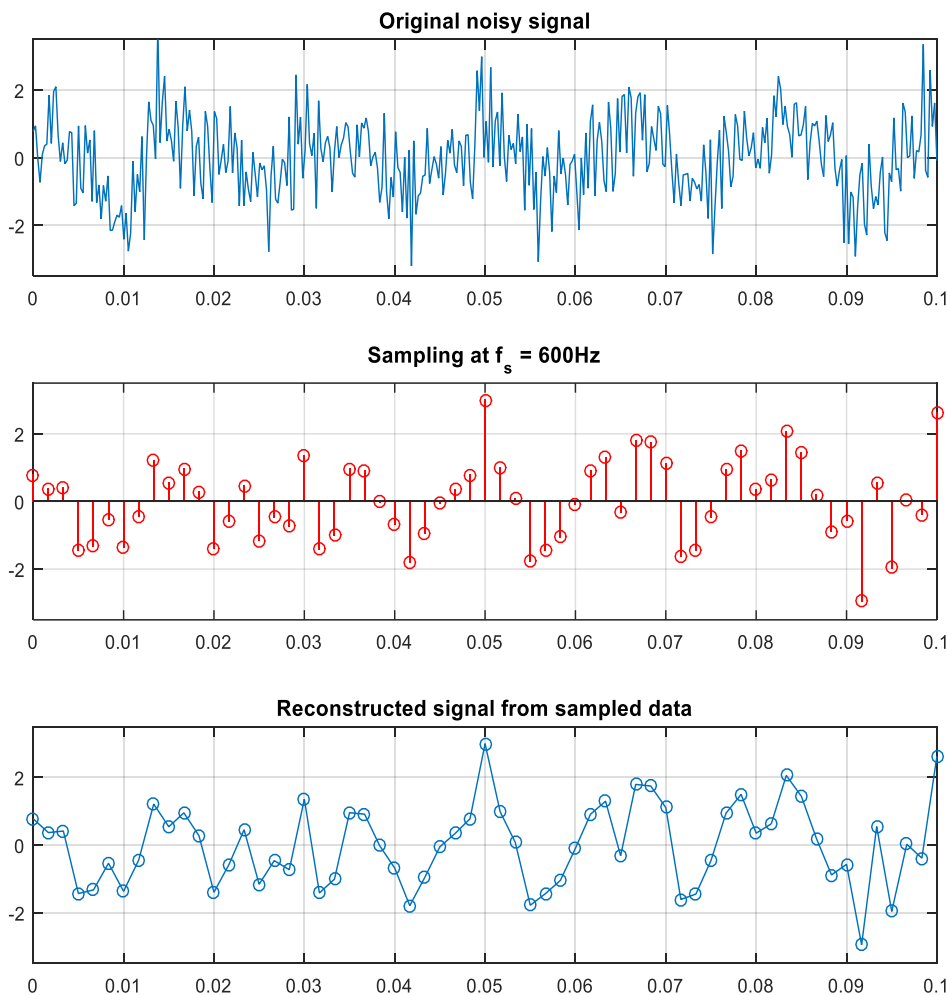
```
z = [y u];  
z = dtrend(z);
```



Input-Output Data Examination

□ Noise Filtering

- High-frequency measurement noise can cause trouble if it is not filtered out before sampling the signals.
- The remedy is to use analog low-pass filters before the signals are sampled.
- Filtering concentrates the identification to the frequency range of interest by increasing the SNR.
- This figures show sampling of noisy data with and without filtering.

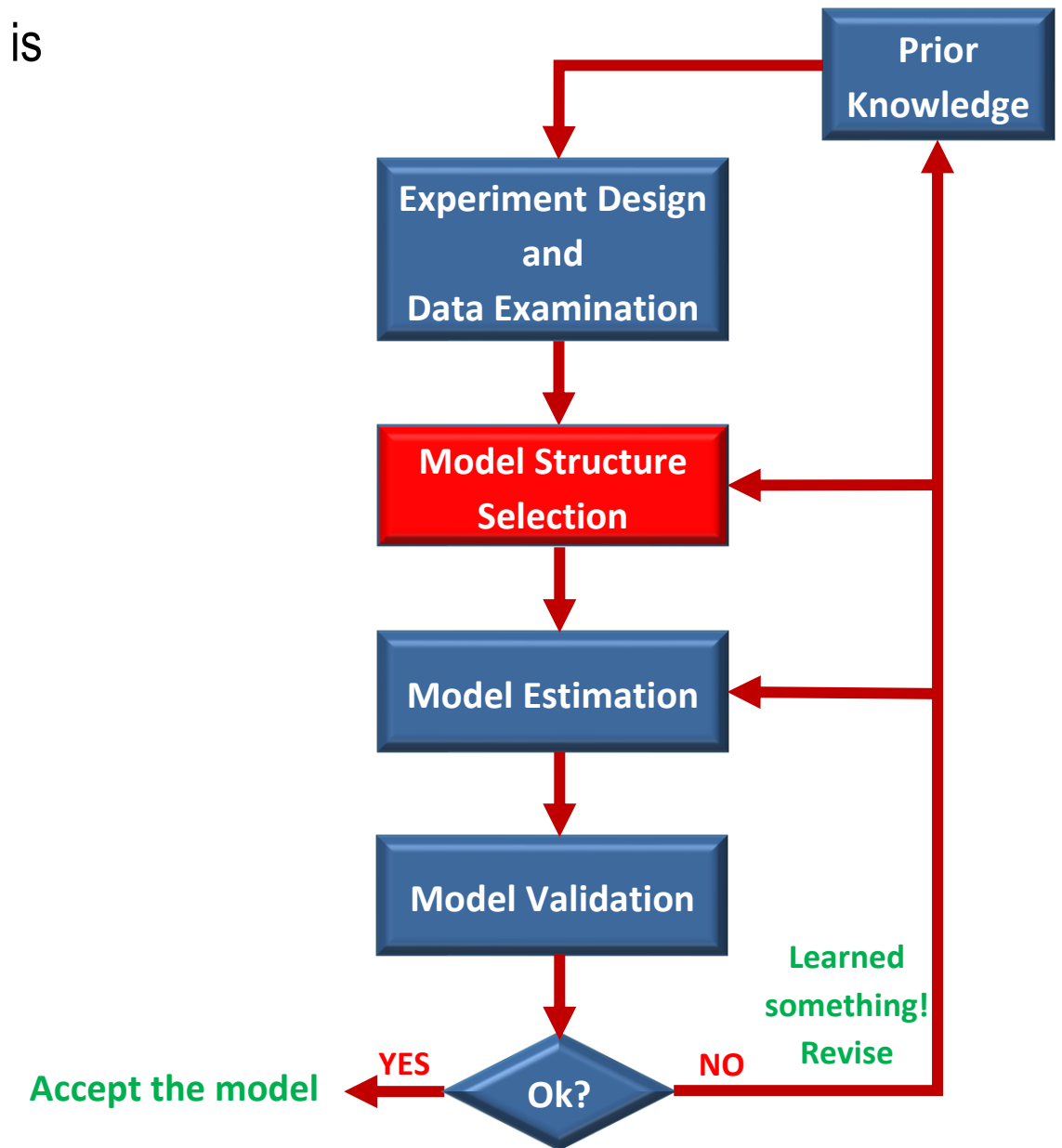


Note that filtering of data may affect the noise model

System Identification Procedure

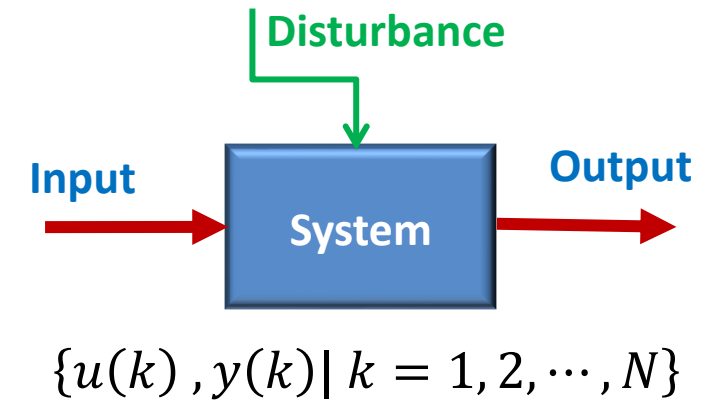
❑ Model Structure Selection

- The **goal** is to determine a **parametric model** for the system. The first step is selecting an **appropriate model structure**.
- **Continuous Time Models**
 - Transfer Function Model
- **Discrete Time Models**
 - Linear Regression Models
 - Transfer Function Models
 - Box-Jenkins (BJ) Model
 - Output-Error (OE) Model
 - ARX Model



Linear Regression Models

- Assume that collected input-output data points of a **system** are available.
- Linear regression model** is the simplest type of a **parametric model**, which the model is **linear** with respect to its **parameters**.



Linear Regression Model

$$y(k) = \boldsymbol{\varphi}^T(k) \boldsymbol{\theta} + e(k), \quad k = 1, \dots, N$$

Output
observations

Regression
vector

Unknown
parameters vector

White noise

where

$y(k)$: the **observed (or measured) data**,

$\boldsymbol{\theta}$: vector of **unknown parameters**,

$\boldsymbol{\varphi}(k)$: is called **regression vector**, which is a vector of **known quantities** that may depend on other known variables. The entries of regression vector are called **regressors**,

$e(k)$: **White noise** with **zero mean** and **known variance of σ_e^2** which is assumed to be **uncorrelated with the regressors**.

$$\text{mean}[e(k)] = 0, \quad \text{var}[e(k)] = \sigma_e^2$$

- The **observations**, $y(k)$, and **regressors** $\boldsymbol{\varphi}(k)$, are obtained from the collected input-output data.

Linear Regression Models

- The **linear regression models** can be used to model the **static systems** and **dynamic systems**.

$$y(k) = \boldsymbol{\varphi}^T(k) \boldsymbol{\theta} + e(k), \quad k = 1, \dots, N$$

- In modeling of **static systems**, the **regressors** depends only on the **current input data**.
For example:

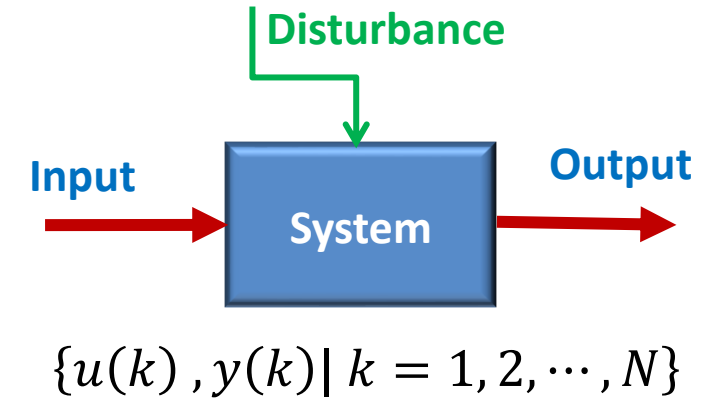
$$y(k) = \theta_0 + \theta_1 u(k) + \theta_2 u(k)^2 + \theta_3 u(k)^3 + e(k)$$

$$y(k) = \underbrace{[1 \quad u(k) \quad u(k)^2 \quad u(k)^3]}_{\boldsymbol{\varphi}^T(k)} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}}_{\boldsymbol{\theta}} + e(k)$$

- In **dynamic systems** modeling, the **regressors** may depend on the **past output** data, and **current and past input** data.
For example:

$$y(k) = \theta_0 y(k-2) + \theta_1 y(k-1) + \theta_2 u(k)^2 + \theta_3 u(k-3) + e(k)$$

$$y(k) = \underbrace{[y(k-2) \quad y(k-1) \quad u(k)^2 \quad u(k-3)]}_{\boldsymbol{\varphi}^T(k)} \underbrace{\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}}_{\boldsymbol{\theta}} + e(k)$$



The models are **linear** in terms of the **parameters** $\boldsymbol{\theta}$ not the regressors $\boldsymbol{\varphi}(k)$.

Linear Regression Modeling of Static Systems

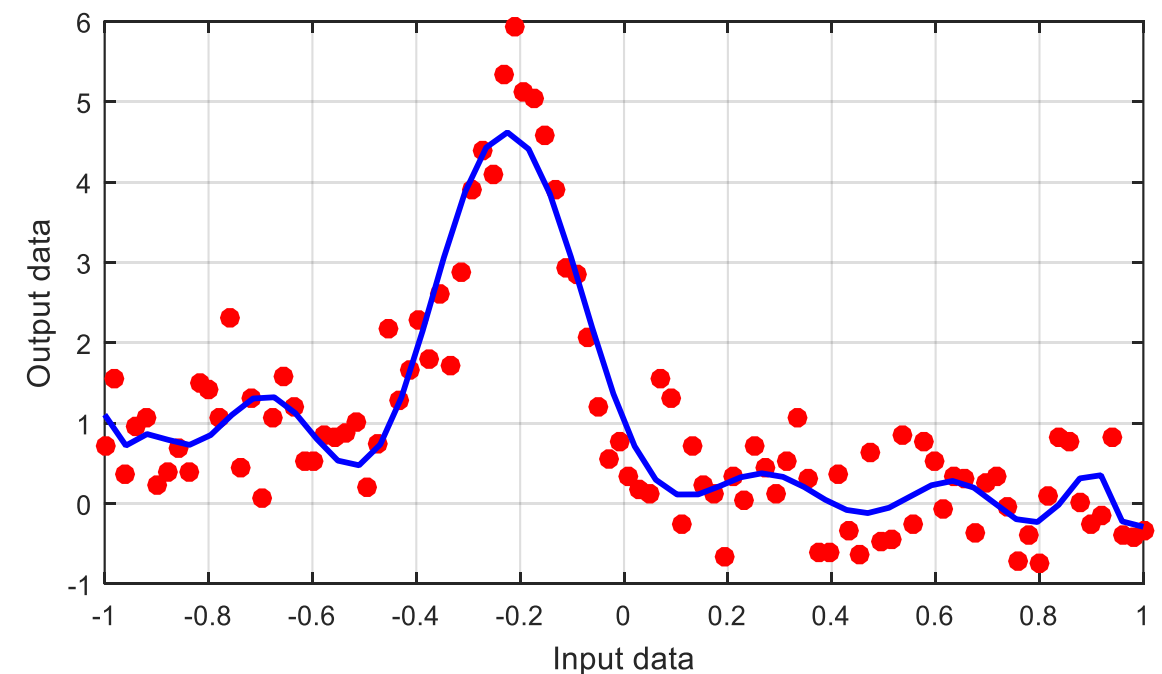
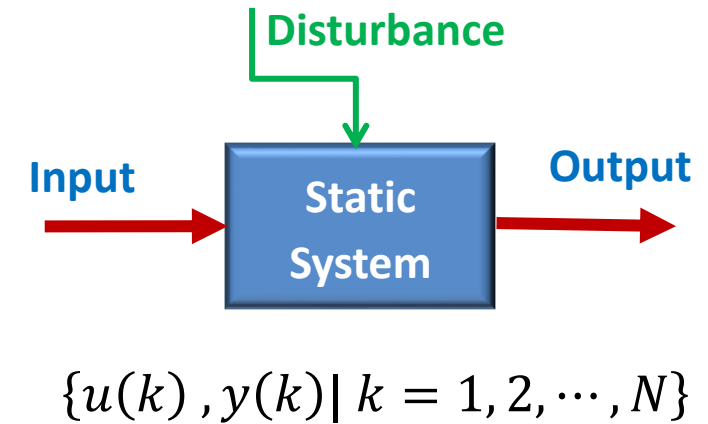
- Assume that collected input-output data points of a **static system** are available.

$$\{u(k), y(k) \mid k = 1, 2, \dots, N\}$$

- Linear regression model** can be used for **curve fitting** in the data points, by modeling the relation between the output variable and the regressors.

$$y(k) = \theta_0 + \theta_1 u(k) + \theta_2 u(k)^2 + \dots + \theta_n u(k)^n + e(k)$$

- In **curve fitting procedure** the goal is to find a **Linear Regression Model** with an **appropriate order** that provides the best fit to the data points by using **Least-squares method**.



Linear Regression Modeling of Static Systems

Linear Regression Model Formulation

- The general form of the linear regression model for static systems is obtained as below

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

- Based on the available dataset ($k = 1, 2, \dots, N$) we have the following set of linear equations

$$\begin{cases} y(1) = \theta_n u(1)^n + \theta_{n-1} u(1)^{n-1} + \dots + \theta_2 u(1)^2 + \theta_1 u(1) + \theta_0 + e(1) \\ y(2) = \theta_n u(2)^n + \theta_{n-1} u(2)^{n-1} + \dots + \theta_2 u(2)^2 + \theta_1 u(2) + \theta_0 + e(2) \\ \vdots \\ y(N) = \theta_n u(N)^n + \theta_{n-1} u(N)^{n-1} + \dots + \theta_2 u(N)^2 + \theta_1 u(N) + \theta_0 + e(N) \end{cases}$$

- We can represent these equations in **matrix-vector** form as below

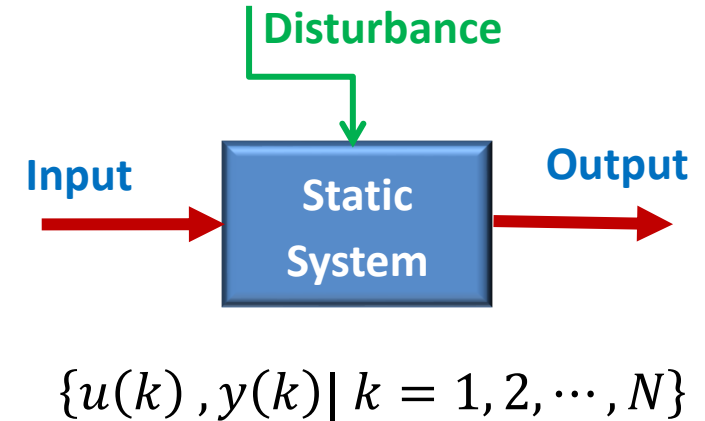
$$\begin{bmatrix} y(1) \\ y(2) \\ \vdots \\ y(N) \end{bmatrix} = \begin{bmatrix} u(1)^n & u(1)^{n-1} & \dots & u(1) & 1 \\ u(2)^n & u(2)^{n-1} & \dots & u(2) & 1 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ u(N)^n & u(N)^{n-1} & \dots & u(N) & 1 \end{bmatrix} \begin{bmatrix} \theta_n \\ \theta_{n-1} \\ \vdots \\ \theta_1 \\ \theta_0 \end{bmatrix} + \begin{bmatrix} e(1) \\ e(2) \\ \vdots \\ e(N) \end{bmatrix}$$

Output observations
 $N \times 1$

Regressors matrix
 $N \times n$

Unknown parameters
 $n \times 1$

White noise
 $N \times 1$



$$\mathbf{Y} = \Phi \boldsymbol{\theta} + \boldsymbol{\epsilon}$$

Overdetermined linear system
No exact solution
The parameters must be estimated

Linear Regression Modeling of Static Systems

□ Least-Squares Estimation Problem

- In **over-determined linear equations**, there is no exact solution, so we resort the equation to solve it in the **Least-Squares** sense and **estimate** the parameters.

$$\mathbf{Y} = \Phi\mathbf{\theta} + \epsilon$$

- Therefore, there are **residuals** or **estimation error** from each of the equations.

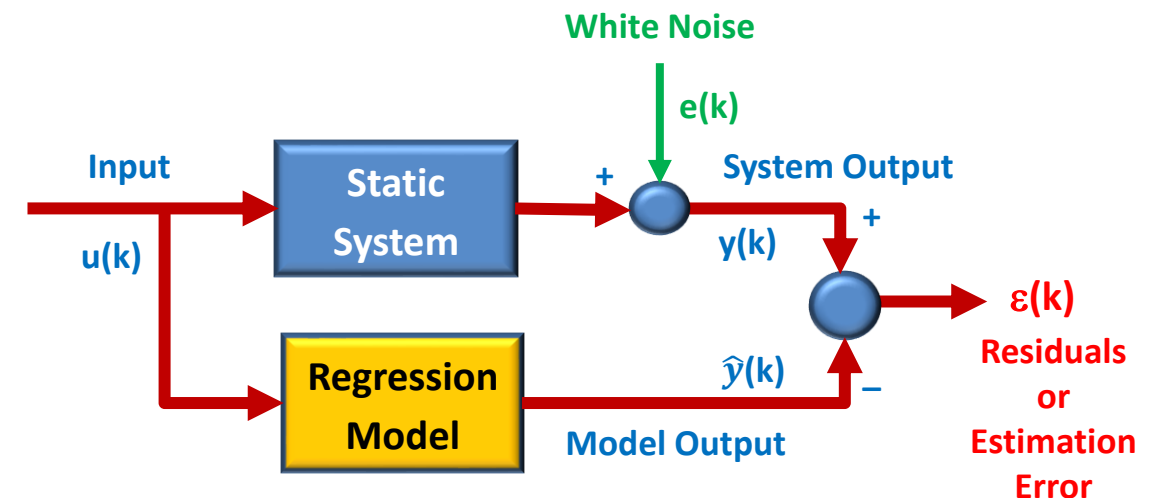
$$\text{Residuals} \rightarrow \boldsymbol{\varepsilon} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \Phi\hat{\boldsymbol{\theta}}$$

where $\hat{\boldsymbol{\theta}}$ is vector of the **estimated variables** and $\boldsymbol{\varepsilon}$ is called the vector of the **residuals**.

- In **Least-Squares Estimation** method the goal is to **minimize** the **Mean Square of the Estimation Error** or the residuals

$$J(\theta) = \frac{1}{N} \sum_{k=1}^N \varepsilon^2(k; \theta) = \frac{1}{N} \|\boldsymbol{\varepsilon}\|_2^2 = \frac{1}{N} \|\mathbf{Y} - \Phi\hat{\boldsymbol{\theta}}\|_2^2$$

MSE loss function



Linear Regression Modeling of Static Systems

□ Least-Squares Estimation Solution

- The unique feature of the MSE loss function is that it can be developed from a **system of linear equations**, and it is a **quadratic function** of unknown parameters. Therefore, it can be minimized **analytically**.

$$J(\theta) = \frac{1}{N} \|\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}}\|_2^2$$

- By expanding the loss function criterion, we have

$$\begin{aligned} \|\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}}\|_2^2 &= (\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}})^T (\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}}) = (\mathbf{Y}^T - \hat{\boldsymbol{\theta}}^T \Phi^T) (\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}}) = \mathbf{Y}^T \mathbf{Y} - \mathbf{Y}^T \Phi \hat{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}^T \Phi^T \mathbf{Y} + \hat{\boldsymbol{\theta}}^T \Phi^T \Phi \hat{\boldsymbol{\theta}} \\ &= \mathbf{Y}^T \mathbf{Y} - (\hat{\boldsymbol{\theta}}^T \Phi^T \mathbf{Y})^T - \hat{\boldsymbol{\theta}}^T \Phi^T \mathbf{Y} + \hat{\boldsymbol{\theta}}^T \Phi^T \Phi \hat{\boldsymbol{\theta}} = \mathbf{Y}^T \mathbf{Y} - 2\hat{\boldsymbol{\theta}}^T \Phi^T \mathbf{Y} + \hat{\boldsymbol{\theta}}^T \Phi^T \Phi \hat{\boldsymbol{\theta}} \quad \text{Quadratic Form} \end{aligned}$$

- Therefore, we can minimize the **loss function** by setting the gradient to zero

$$\frac{d}{d\hat{\boldsymbol{\theta}}} \|\mathbf{Y} - \Phi \hat{\boldsymbol{\theta}}\|_2^2 = 0 \quad \longrightarrow \quad \frac{d}{d\hat{\boldsymbol{\theta}}} [\mathbf{Y}^T \mathbf{Y} - 2\hat{\boldsymbol{\theta}}^T \Phi^T \mathbf{Y} + \hat{\boldsymbol{\theta}}^T \Phi^T \Phi \hat{\boldsymbol{\theta}}] = 0$$

$$-2\Phi^T \mathbf{Y} + 2\Phi^T \Phi \hat{\boldsymbol{\theta}} = 0 \quad \rightarrow \quad \boxed{\Phi^T \mathbf{Y} = \Phi^T \Phi \hat{\boldsymbol{\theta}}} \quad \text{Normal Equation}$$

- Solve the **Normal Equation** to find the **estimated parameters** $\hat{\boldsymbol{\theta}}$ that minimize the MSE loss function:

$$\text{Estimated Parameters} \rightarrow \boxed{\hat{\boldsymbol{\theta}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{Y} = \Phi^\# \mathbf{Y}}$$

Pseudo Inverse

Derivatives of Quadratic Forms

$$\frac{\partial \mathbf{x}^T B \mathbf{x}}{\partial \mathbf{x}} = (B + B^T) \mathbf{x}$$

If B is a symmetric matrix we have

$$\frac{\partial \mathbf{x}^T B \mathbf{x}}{\partial \mathbf{x}} = 2B \mathbf{x}$$

Linear Regression Modeling of Static Systems

□ Model Order Selection

- In the **linear regression modeling** the important part is appropriate selecting the **model order (n)** or the **number of unknown parameters θ** .
- Assume that the estimated models based on the collected N samples of data for different model orders are available:

$$n = 1 \rightarrow \hat{y}(k) = \theta_1 u(k) + \theta_0$$

$$n = 2 \rightarrow \hat{y}(k) = \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0$$

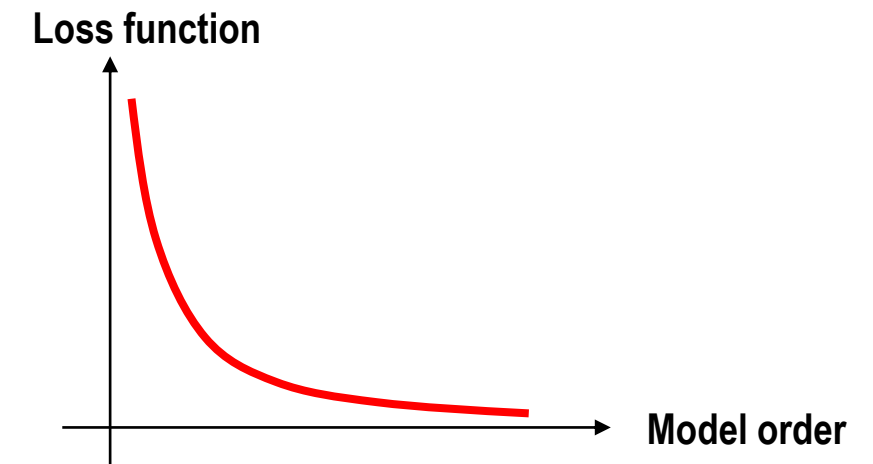
$$n = 3 \rightarrow \hat{y}(k) = \theta_3 u(k)^3 + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0$$

$\vdots$

- The **best order** can be determined by evaluating the **loss function** over a range of orders, for example $n = 1, 2, \dots, 10$ and selecting the best order which minimizes the loss function.

$$\text{Loss Function} \rightarrow J(\theta) = \frac{1}{N} \sum_{k=1}^N \varepsilon^2(k; \theta) = \frac{1}{N} \sum_{k=1}^N (y(k) - \hat{y}(k; \theta))^2$$

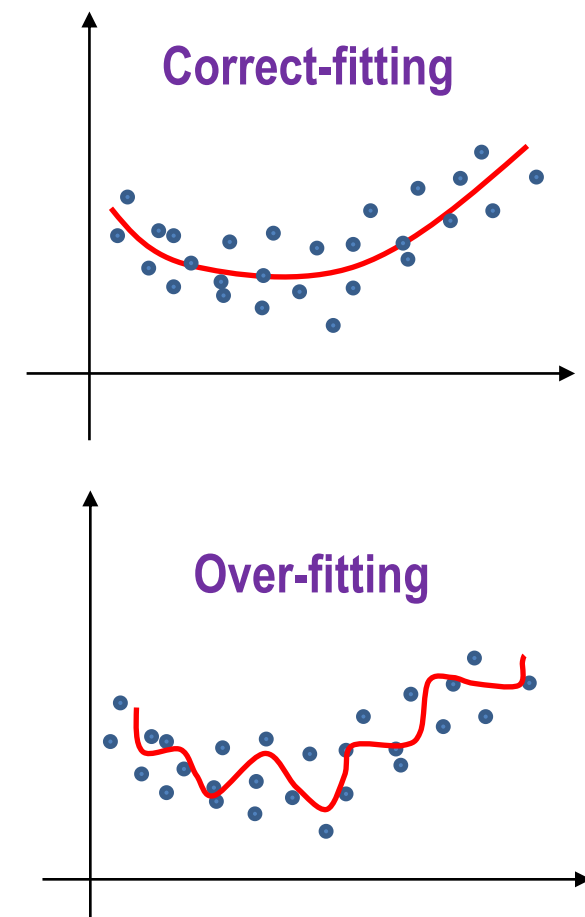
- In general, by **increasing the order of model** (number of parameters) the **Least-Squares Estimation error** and the **loss function** value **will decrease monotonically**.
- However, it does not mean to have a good estimation, because it makes the **model more complex** than we need and causes the **Over-fitting to the data**.



Linear Regression Modeling of Static Systems

□ Model Order Selection

- Typically, when the model order be selected too complex, it actually models the **noise dynamic** not the true system. This issue is called **Over-fitting**.
- In the case of **over-fitting** if we validate the estimated model by the **same identification data** the result looks **good**, but if we use **new fresh dataset for validation** the results will be **poor**.
- To **avoid the over-fitting** we have to split the input-output dataset Z (N samples) in **two datasets**:
 - Identification dataset $\rightarrow Z_{id}$
 - Validation dataset $\rightarrow Z_{val}$
- Then by applying this technique we can determine the **optimum order** of the given data and compute the **quality of fit** of the estimated model by validating the model on a dataset where neither it was estimated.
- Note that if the number of available samples of data is **not enough** to split equally, we can consider a **larger set of data for identification** and a smaller set for **validation**, and determine the order based on the **validation dataset**.

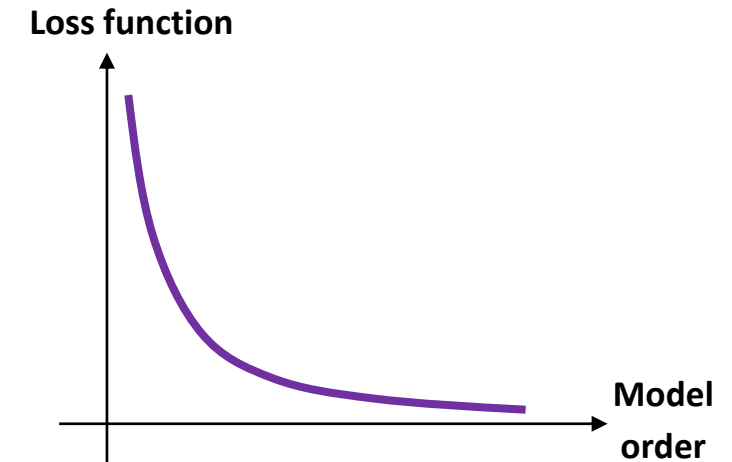


Linear Regression Modeling of Static Systems

□ Model Order Selection

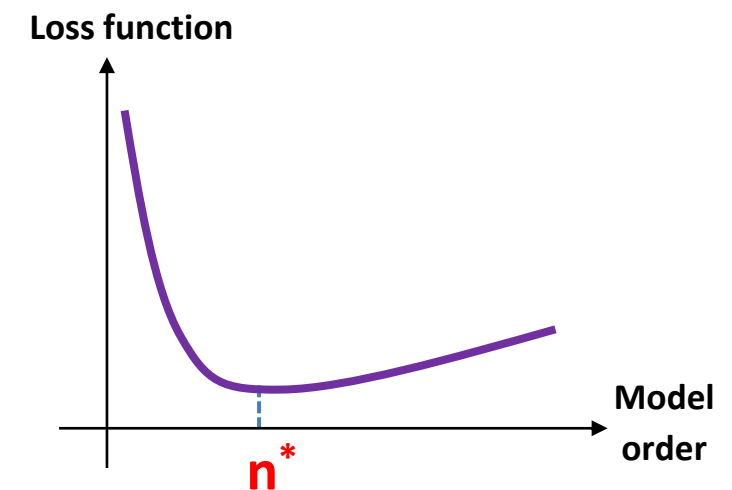
- If we evaluate the **loss function** using the **identification data**, by increasing the order of the model the **Least-squares estimation error** and the **Loss function** value decreases.

$$\text{Loss function using ID data} \rightarrow J(\theta, Z_{id}) = \frac{1}{N_{id}} \sum_{i=1}^{N_{id}} (y(k) - \hat{y}(k; \theta))^2$$



- On the other hand, the **loss function** has a different behavior when we apply it on the **new fresh data**. It **starts to increase** when the model complexity (order) becomes excessive (over-fitting data).

$$\text{Loss function using new data} \rightarrow J(\theta, Z_{val}) = \frac{1}{N_{val}} \sum_{i=1}^{N_{val}} (y(k) - \hat{y}(k; \theta))^2$$



- The **optimum order (n^*)** can be selected by minimizing the loss function using the **new fresh dataset (a dataset other than the identification data)** over a range of different orders (e.g. $n = 1, 2, \dots, 10$) and selecting the best result for minimization.

Linear Regression Modeling of Static Systems

Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

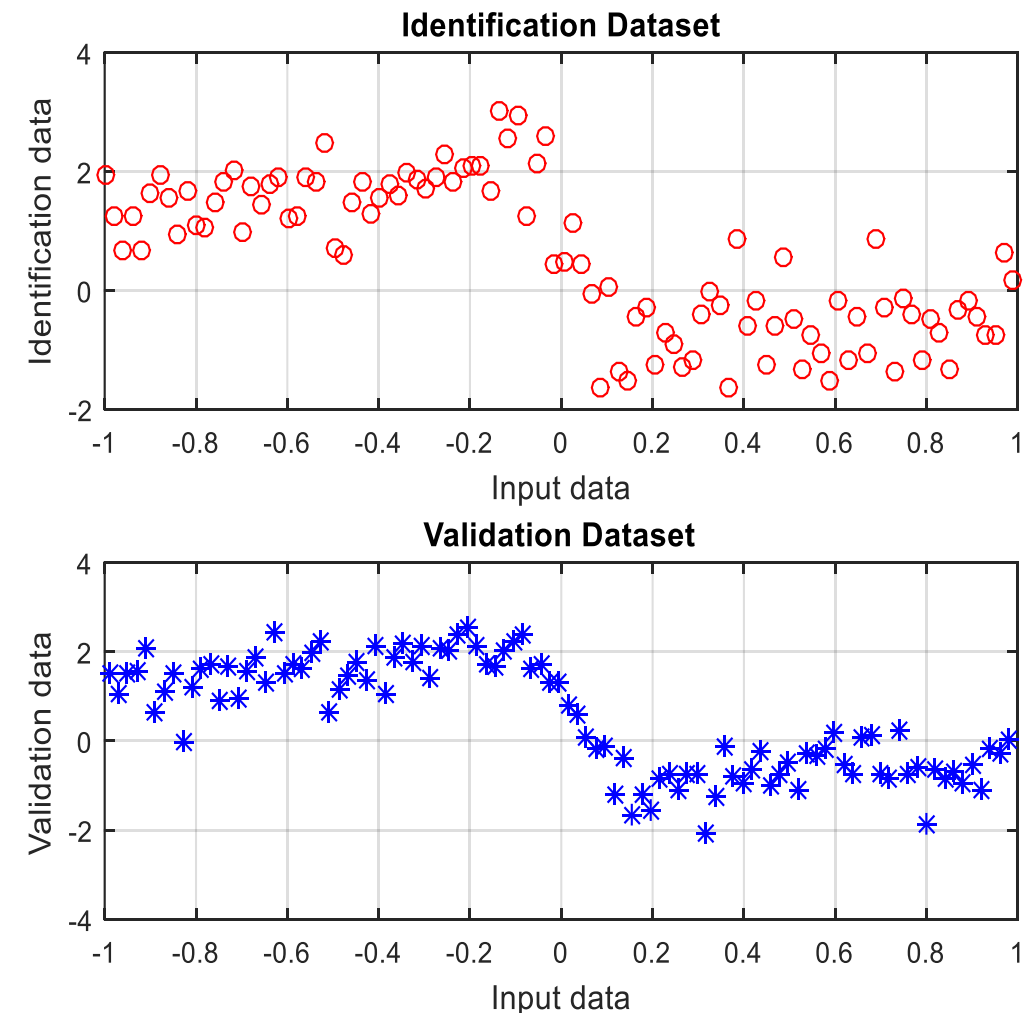
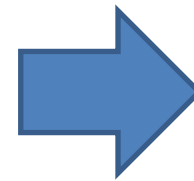
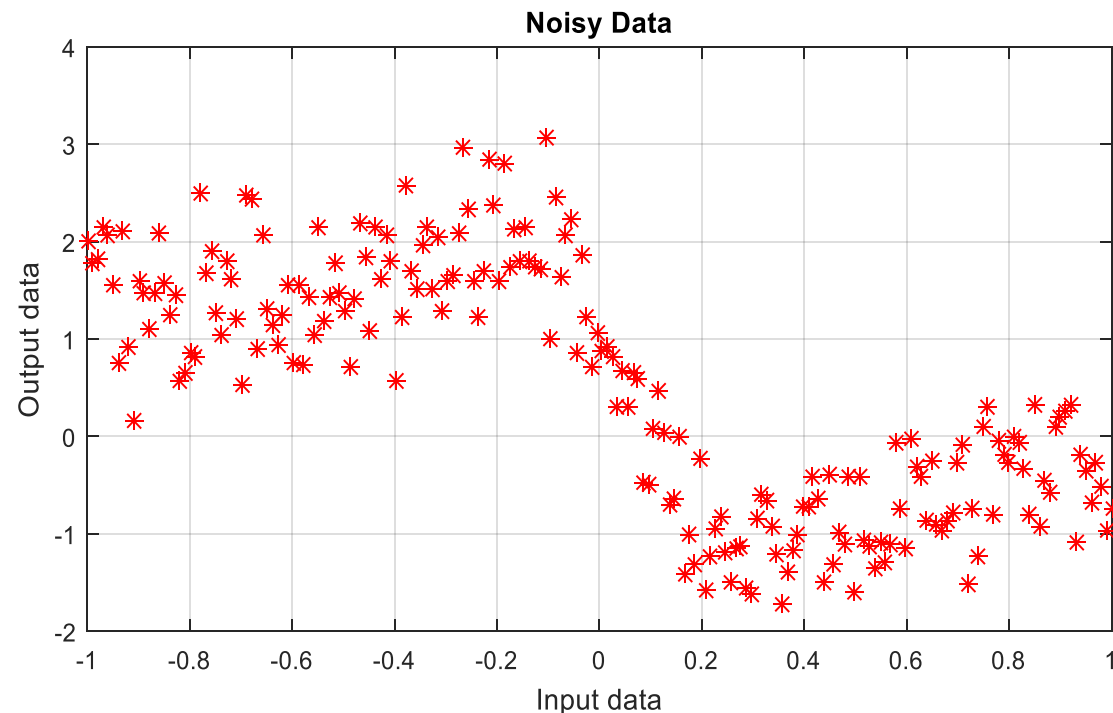
$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

```
plot(u,y, '* r')      ← Plot I/O noisy data (u and y are column vectors)
```

First, split the data to **identification** and **validation** datasets.

```
plot(uid,yid, 'o r')  ← Plot I/O identification data  
plot(uval,yval, '* b') ← Plot I/O validation data
```

$$N_{id} = 100, \quad N_{val} = 100$$



Linear Regression Modeling of Static Systems

Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

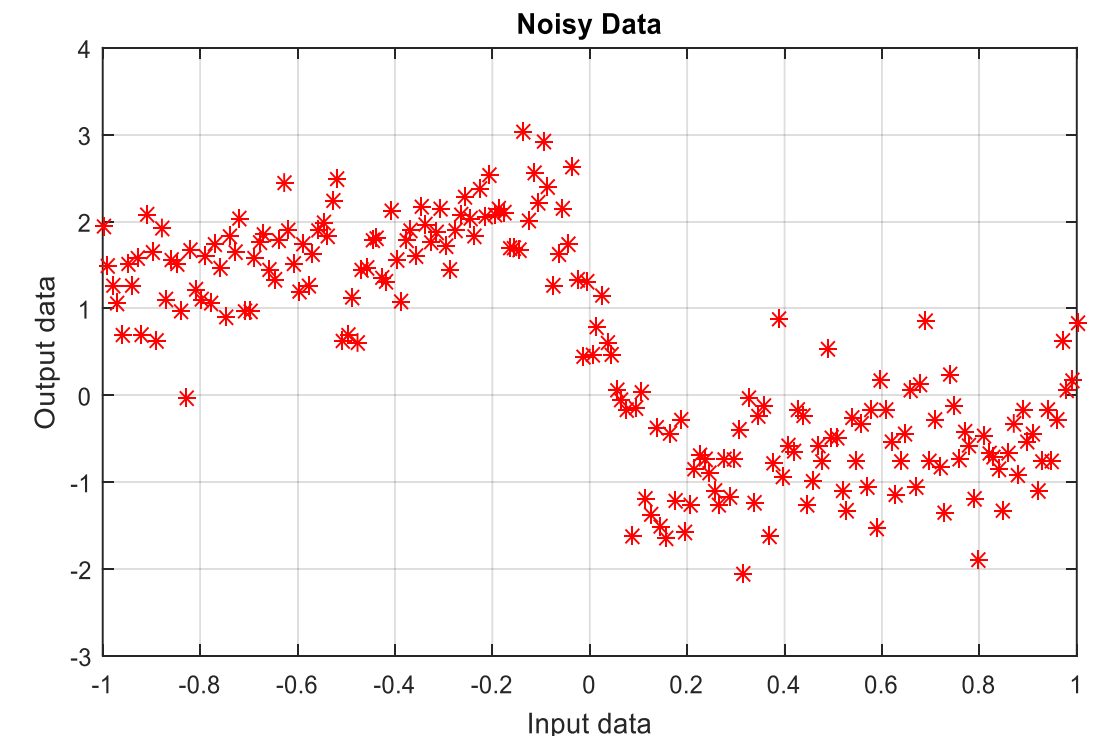
Next, formulate the data in the **Linear Regression** form using the available samples of the **identification data**:

$$\begin{bmatrix} y(1) \\ y(2) \\ \vdots \\ y(N_{id}) \end{bmatrix} = \begin{bmatrix} u(1)^n & u(1)^{n-1} & \dots & u(1) & 1 \\ u(2)^n & u(2)^{n-1} & \dots & u(2) & 1 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ u(N_{id})^n & u(N_{id})^{n-1} & \dots & u(N_{id}) & 1 \end{bmatrix} \begin{bmatrix} \theta_n \\ \theta_{n-1} \\ \vdots \\ \theta_1 \\ \theta_0 \end{bmatrix} + \begin{bmatrix} e(1) \\ e(2) \\ \vdots \\ e(N_{id}) \end{bmatrix} \rightarrow \boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}}$$

$$\text{LS Estimate} \rightarrow \hat{\boldsymbol{\theta}} = \mathbf{X}^\# \mathbf{Y}$$

We can use the following MATLAB code to formulate the linear regression model.

```
Yid = yid;           ← Vector of output observations
Xid = zeros(Nid,n+1); ← Initialize matrix of regressors
for i=1:n+1
    Xid(:,i) = uid.^(n+1-i); ← Generate matrix of regressors
end
theta_hat = pinv(Xid)*Yid; ← LS estimated parameters
```



Linear Regression Modeling of Static Systems

Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

Since, the **model order (n) is unknown**, we consider polynomials with different orders to fit the data:

$$n = 1 \rightarrow \hat{y}(k) = \theta_1 u(k) + \theta_0$$

$$n = 2 \rightarrow \hat{y}(k) = \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0$$

$$n = 3 \rightarrow \hat{y}(k) = \theta_3 u(k)^3 + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0$$

$\vdots$

Model estimation for $n = 5, 10, 20$ and 40 are shown:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\theta} \rightarrow \hat{\boldsymbol{\theta}} = \mathbf{X}^\# \mathbf{Y} \rightarrow \hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\theta}}$$

```
Yid = yid;                                ← Vector of output observations
for n=[5 10 20 40];                        ← Model order
    Xid = zeros(Nid,n+1);                  ← Initialize matrix of regressors
    for i=1:n+1
        Xid(:,i) = uid.^(n+1-i);          ← Generate matrix of regressors
    end
    theta_hat = pinv(Xid)*Yid;              ← LS estimated parameters
    Y_hat = zeros(size(Yid));              ← Initialize estimated output
    Y_hat = Xid*theta_hat;                 ← Estimated y
    plot(u,Y_hat), hold on
end
```


Linear Regression Modeling of Static Systems

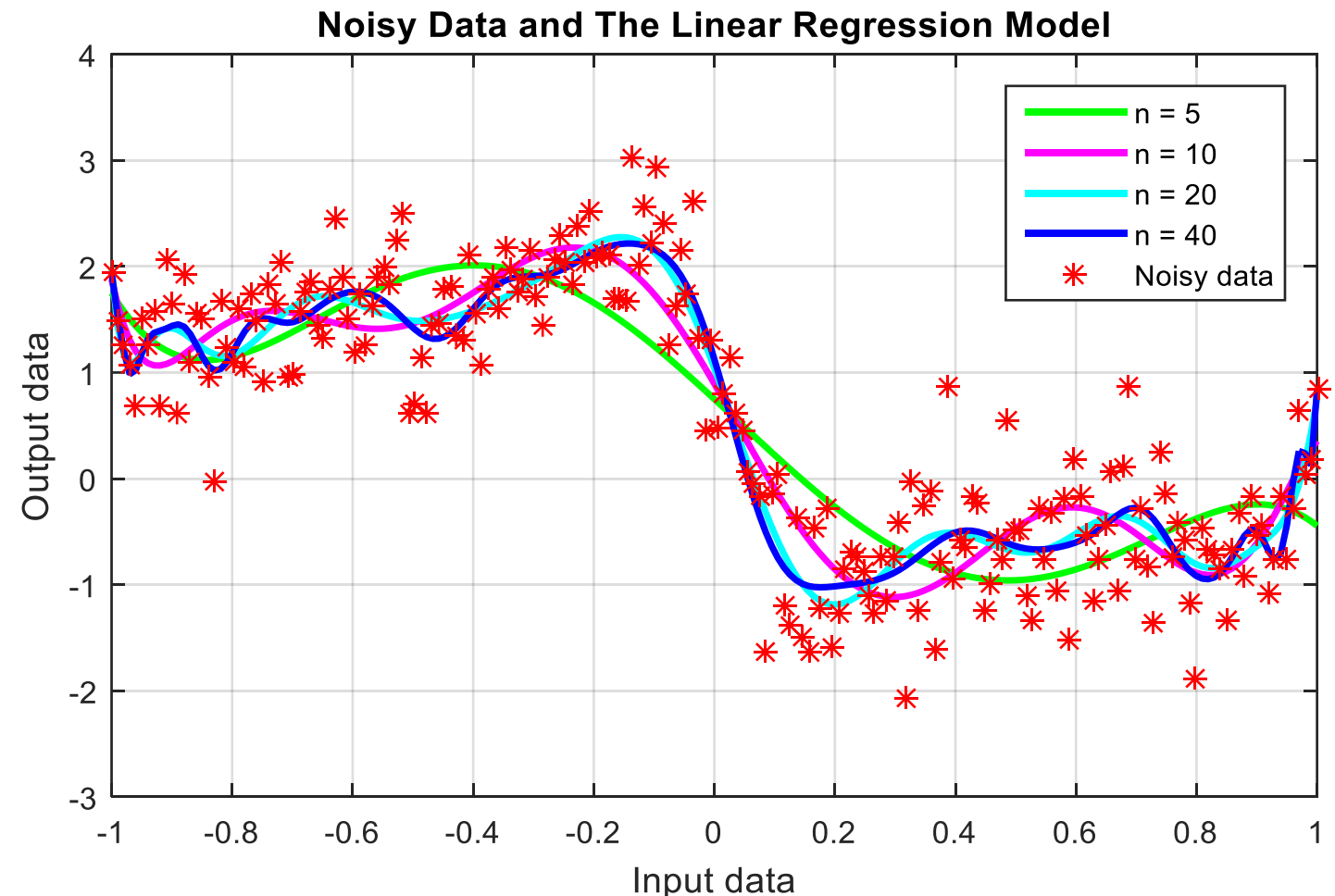
Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \cdots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

Model estimation for $n = 5, 10, 20$ and 40 are shown in the following figures.

- We can see that by increasing the order of polynomial the model starts **over-fitting** the data.
- It means that instead of modeling the **true system**, the model tries to **model the noise**.



Linear Regression Modeling of Static Systems

Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

The optimum order can be selected by minimizing the loss function using the validation dataset.

$$J(\theta, Z_{val}) = \frac{1}{N_{val}} \sum_{i=1}^{N_{val}} (y(k) - \hat{y}(k; \theta))^2$$

Loss function for VAL data

```
Yid = yid; Yval = yval;           ← Vector of output observations
for n=1:20;                       ← Range of the model order
    Xid = zeros(Nid,n+1);         ← Initialize matrix of regressors for ID data
    for i=1:n+1
        Xid(:,i) = uid.^(n+1-i); ← Generate matrix of the regressors for ID data
    end
    theta_hat = pinv(Xid)*Yid;     ← LS estimated parameters using ID data

    Xval = zeros(Nval,n+1);       ← Initialize matrix of the regressors for VAL data
    for i=1:n+1
        Xval(:,i) = uval.^(n+1-i); ← Generate matrix of the regressors for VAL data
    end
    Y_hat = zeros(size(yval));    ← Initialize estimated output
    Y_hat = Xval*theta_hat;       ← Estimated y

    J(n) = immse(yval,Y_hat);     ← MSE Loss function
end
plot(J, 'o- r')
```

$$\mathbf{Y}_{id} = \mathbf{X}_{id} \boldsymbol{\theta} \rightarrow \hat{\boldsymbol{\theta}} = \mathbf{X}_{id}^{\#} \mathbf{Y}_{id}$$
$$\hat{\mathbf{Y}}_{val} = \mathbf{X}_{val} \hat{\boldsymbol{\theta}}$$

Linear Regression Modeling of Static Systems

Example 3

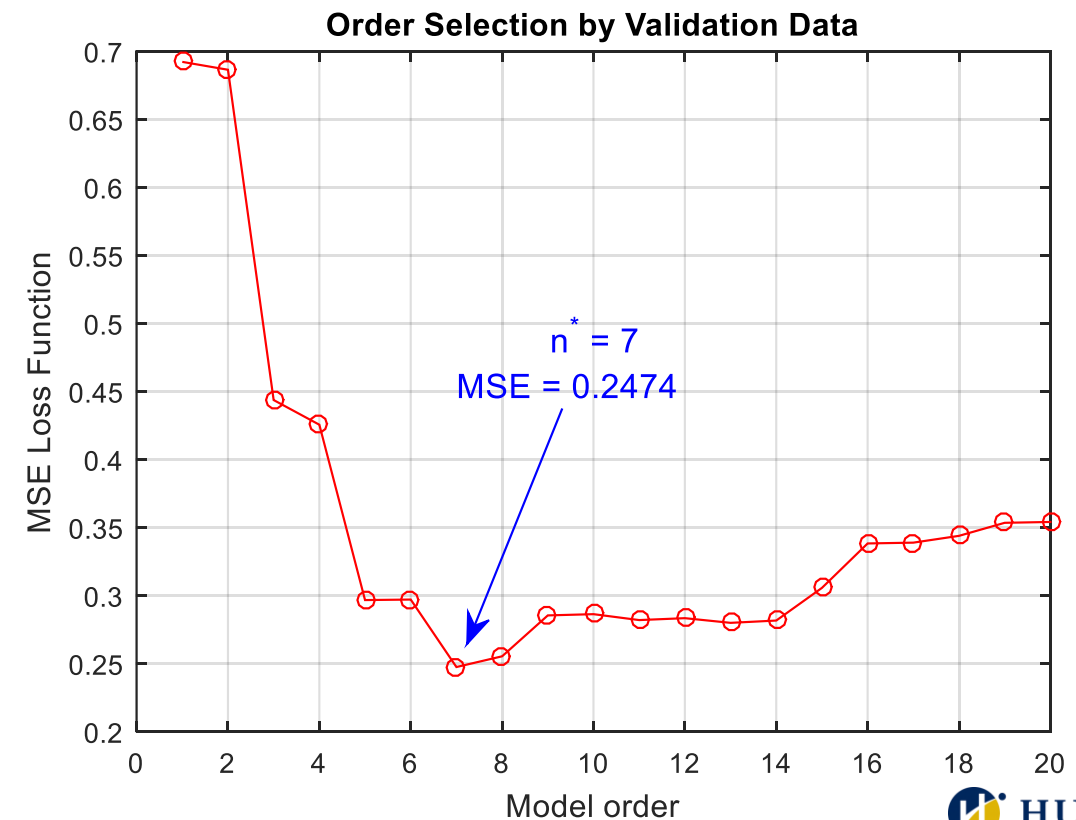
Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

Following figures compare the loss function evaluation for given range of model order $n = 1, 2, \dots, 20$, using the [validation dataset](#) and the [identification dataset](#).

$$J(\theta, Z_{id}) = \frac{1}{N_{id}} \sum_{i=1}^{N_{id}} (y(k) - \hat{y}(k; \theta))^2$$

$$J(\theta, Z_{val}) = \frac{1}{N_{val}} \sum_{i=1}^{N_{val}} (y(k) - \hat{y}(k; \theta))^2$$



Linear Regression Modeling of Static Systems

Example 3

Assume that the following collected $N = 200$ samples of a noisy data is available. Estimate a polynomial model by Least-Squares Estimation method.

$$y(k) = \theta_n u(k)^n + \theta_{n-1} u(k)^{n-1} + \dots + \theta_2 u(k)^2 + \theta_1 u(k) + \theta_0 + e(k)$$

Evaluation results show that the optimum order can be selected as

Optimum order $\rightarrow n^* = 7$

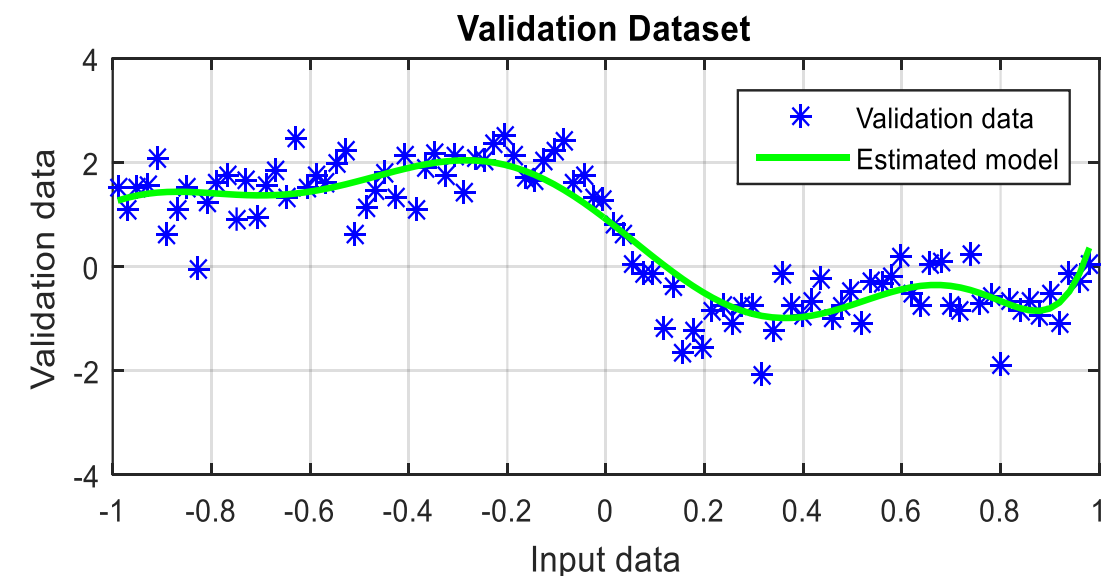
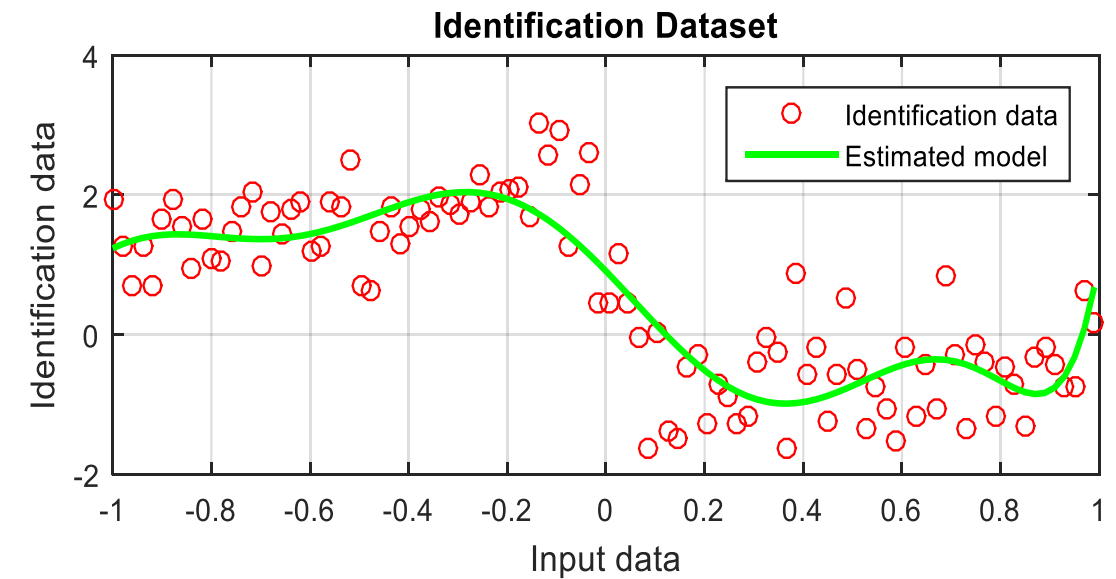
and the loss function evaluation is $MSE = 0.2474$

As a result, the best model structure to estimate the given data will be as

$$y = \theta_7 u^7 + \theta_6 u^6 + \theta_5 u^5 + \theta_4 u^4 + \theta_3 u^3 + \theta_2 u^2 + \theta_1 u + \theta_0$$

Estimated Parameters $\rightarrow$

$$\begin{bmatrix} \theta_7 \\ \theta_6 \\ \theta_5 \\ \theta_4 \\ \theta_3 \\ \theta_2 \\ \theta_1 \\ \theta_0 \end{bmatrix} = \begin{bmatrix} 16.9586 \\ -29.9632 \\ -38.2327 \\ 22.8056 \\ 27.5786 \\ -6.1394 \\ -7.3441 \\ 0.8196 \end{bmatrix}$$



THANK YOU